

SAT Math Desk Packet

Student: Karina (Newton North High School) **Instructor desk copy** — Digital SAT Math, ~680 → ~780 **Cadence:** Saturday 4 hours + Sunday 4 hours **Compiled from** the sat -prep/ folder in Karina_CommonApp

This is one printable packet of everything in the SAT prep repository. It is not condensed. Use it at the desk: point, write, and talk.

PSDA = Problem-Solving and Data Analysis (ratios, percents, data, probability, statistical claims).
About 15% of SAT Math.

How to use this printout

1. Weekend 1 Saturday is a high-level Q&A tour of **all four** domains. Student talks first.
2. Weekends 2–4 go deep on Advanced Math, then PSDA, then Geometry & Trig.
3. Practice sets are mixed into each weekend. Time them when the script says to.
4. **Answer keys start after Weekend 4.** Skip printing those pages if Karina should not see solutions during a set. Or fold them under / keep them on the instructor side of the desk.
5. Official Bluebook / Question Bank items are **not** copied here. Links are in the resources section.

Packet contents

1. Digital SAT overview
2. Math content map (skill checklist)
3. Curriculum roadmap
4. Official resources
5. Preparation plan
6. Weekend 1 — all four domains (Sat / Sun / practice / optional Algebra drill)
7. Weekend 2 — Advanced Math
8. Weekend 3 — PSDA
9. Weekend 4 — Geometry & Trig + Module 1

Digital SAT overview

Audience: Instructor + student (Karina)
Cycle: Digital SAT (Bluebook) — current College Board format
Last updated: 2026-08-20

Part of sat -prep/ only. Not related to the Common App essay packet in docs/.

1. Big picture

The SAT is now **fully digital** (U.S. since spring 2024). Students take it in College Board’s **Bluebook** app.

Section	Modules	Time	Questions	Score
Reading and Writing	2 × 32 min	64 min	54 (incl. pretest)	200–800
Math	2 × 35 min	70 min	44 (incl. pretest)	200–800
Break	between sections	10 min	—	—
Total		~2 hr 14 min		400–1600

Your focus: Math ~680 → ~780.

2. What changed vs the old paper SAT (important for instructors who took it ~5–6 years ago)

Old paper SAT (through ~2023)	Digital SAT (now)
~3 hours	~2 hours 14 minutes
Separate No Calculator + Calculator Math	Calculator allowed on all Math
Linear form (same difficulty for everyone)	Multistage adaptive (Module 2 depends on Module 1)
Longer wordy stems	Shorter, more focused questions
Grid-in answers on paper	Student-produced response (SPR) in Bluebook
Bring your own calc	Desmos graphing calculator built into Bluebook (+ optional personal approved calc)
Domains: Heart of Algebra, Passport to Advanced Math, etc.	Renamed/refined: Algebra, Advanced Math, PSDA, Geometry & Trigonometry

Takeaway for teaching: do **not** drill “no-calculator tricks” as a separate half of Math. Teach **Desmos fluency** as a core skill.

3. How Math adaptive modules work

1. **Module 1:** Mix of easy / medium / hard (routing module).
2. Performance on Module 1 routes the student to a **harder** or **easier** Module 2.
3. Within a module, questions generally go **easier** → **harder**.
4. You can skip/flag and move freely **inside** a module; you **cannot** go back after the module timer ends.

Score implication: Missing easy Module 1 items can send her to the easier Module 2 and **cap** upside. Accuracy on the first ~10–12 “should-get” Algebra items matters a lot for a 780 push.

4. Math question formats

- **Multiple choice (~75%):** 4 options (A–D).
- **Student-produced response / SPR (~25%):** Type the answer (integer, decimal, or fraction per Bluebook rules). No choices — calculation care matters.

About **30%** of Math questions are **in context** (science, social studies, or real-world scenarios).

Shorter than old SAT word problems, but still require translating English → math.

5. Four Math content domains (College Board)

Domain	Approx. %	# questions	What it tests
Algebra	~35%	13–15	Linear equations, inequalities, systems, linear functions
Advanced Math	~35%	13–15	Nonlinear: quadratic, exponential, absolute value, polynomials, rationals, radicals
Problem-Solving & Data Analysis (PSDA)	~15%	5–7	Ratios, %, units, data displays, probability, statistical claims

PSDA is not a separate school subject. It is College Board’s label for the “story + table + percent + probability + what can you conclude” questions. Only 5–7 items, but they are easy points to drop on wording. | **Geometry & Trigonometry** | ~15% | 5–7 | Area/volume, angles/triangles, right-triangle trig, circles |

Every module includes questions from **all four** domains.

Full skill checklist: 01-math-content-map.md.

6. Pacing target

- Math: **70 minutes / 44 questions** $\approx$ **1.6 minutes per question**.
 - Strategy for ~680 $\rightarrow$ 780: 1. Bank easy Algebra early (high accuracy). 2. Use Desmos for graphs, systems, and “which equation matches” checks. 3. Don’t bleed 3+ minutes on one hard Advanced Math item before Module 1 is secure. 4. Always attempt SPRs — blank SPR is free miss.
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7. Tools to master

1. **Bluebook** — practice tests + UI (mark for review, answer eliminator, SPR entry).
2. **Desmos (in Bluebook)** — graph lines/parabolas, solve systems visually, check intersections.
3. **Official Student Question Bank** — filter by domain / skill / difficulty.
4. **Error log** — every miss: skill tag + mistake type (concept / setup / arithmetic / rushed).

Links: `resources.md`.

8. Score goal framing

Score band	Rough meaning for Math
~680	Solid Algebra base, inconsistent Advanced Math / careless Module 1 misses
~730	Strong Algebra + improving Advanced Math
~780 (target)	High accuracy on Algebra + Advanced Math; few PSDA/Geometry freebies left on table

Path in this folder: Weekend 1 four-domain survey $\rightarrow$ Weekends 2–4 deep nitpick (B, then PSDA, then Geometry) $\rightarrow$ timed Bluebook Math modules $\rightarrow$ weak-skill cram.

Math content map

Checklist for Karina. Mark ☐ / ☐ / ☒ as you teach.

Sources: College Board Digital SAT Math domains / skill lists (Algebra, Advanced Math, PSDA, Geometry & Trigonometry).

Domain A — Algebra (~35%, 13–15 questions)

#	Skill	Status	Notes
A1	Linear equations in one variable	☐	Isolate; clear fractions/decimals
A2	Linear equations in two variables	☐	Forms: slope-intercept, standard, point-slope
A3	Linear functions	☐	Rate of change; intercept meaning in context
A4	Systems of two linear equations in two variables	☐	Substitution, elimination, Desmos intersection
A5	Linear inequalities in one or two variables	☐	Solution sets; which values satisfy

Weekend 1: high-level Q&A with the other three domains. **Keep warm** every weekend.

Domain B — Advanced Math (~35%, 13–15 questions)

#	Skill	Status	Notes
B1	Equivalent expressions	☐	Factor, expand, rewrite radicals/exponents
B2	Nonlinear equations in one variable	☐	Quadratic, absolute value, radical, rational, exponential

#	Skill	Status	Notes
B3	Systems involving nonlinear equations	☐	Line + parabola, etc.
B4	Nonlinear functions	☐	Quadratic/exponential graphs & parameters

Weekend 1: survey (quadratics, exponentials, abs value, nonlinear systems).

Weekend 2: full B deep dive.

Domain C — Problem-Solving & Data Analysis (~15%, 5–7 questions)

#	Skill	Status	Notes
C1	Ratios, rates, proportional relationships, and units	☐	Unit analysis
C2	Percentages	☐	% change, successive %
C3	One-variable data: distributions; center & spread	☐	Mean/median/range; histograms
C4	Two-variable data: models & scatterplots	☐	Linear vs exponential fit
C5	Probability and conditional probability	☐	Tables / two-way frequency
C6	Inference from sample statistics & margin of error	☐	Conceptual
C7	Evaluating statistical claims (studies/experiments)	☐	Bias, confounding (light)

Weekend 1: survey. **Weekend 3:** full PSDA deep dive.

Domain D — Geometry & Trigonometry (~15%, 5–7 questions)

#	Skill	Status	Notes
D1	Area and volume	☐	Formulas; scale factors

#	Skill	Status	Notes
D2	Lines, angles, and triangles	☐	Parallel lines, similarity, congruence
D3	Right triangles and trigonometry	☐	Pythagorean; sin/cos/tan; special rights
D4	Circles	☐	Arc, sector, equation of circle (as tested)

Weekend 1: survey. **Weekend 4:** full Geometry & Trig + Module 1 mix.

Cross-cutting habits (score killers at ~680)

Habit	Status
Translate word problem → equation before calculating	☐
Check SPR entry format (sign, fraction, decimal)	☐
Desmos verification for graphs/systems	☐
Module 1: do not miss “easy” Algebra	☐
Error log after every practice set	☐

Curriculum roadmap

Student: Karina · **Start Math:** ~680 · **Target:** ~780

Cadence: Saturday 4h + Sunday 4h

Model: She already knows the school-math. We **survey, nitpick, then drill** — not teach from zero.

PSDA = Problem-Solving and Data Analysis (ratios, percents, data, probability, statistical claims).

Assumptions (update when confirmed): practice Math $\approx$ 680; exact RW/total TBD.

Weekend map

Weekend	Folder	Theme	Domains
1	weekend-01/	All four domains , high-level Q&A + mixed diagnostic	A · B · C · D
2	weekend-02/	Advanced Math deep dive (traps + volume)	B1–B4
3	weekend-03/	PSDA wording / tables / claims	C1–C7
4	weekend-04/	Geometry & Trig + Module 1 pacing	D1–D4 + mixed
5	<i>(create when ready)</i>	Full Bluebook Math + error clinic	Mixed
6+	<i>(create when ready)</i>	Weak-skill cram + timed modules	Targeted

Weekend 1 Saturday is a **tour of all four** (one ~50 min block each). Weekends 2–4 go deep on B, then C, then D, while keeping mixed items so the other domains stay warm.

Score checkpoints

After weekend	Suggested check
1	Mixed 16-item set (all four domains) $\geq 85\%$ untimed-quality, timed 24 min
2	Advanced Math practice set $\geq 80\%$
3	PSDA practice set $\geq 80\%$
4	Geometry set $\geq 80\%$ + Module 1 sim ≤ 2 misses
5	Full Bluebook Math section; log every miss by skill ID

Weekly between-weekend homework (default)

1. Official Student Question Bank: **20–30** filtered items on that weekend’s skills.
2. Update 01-math-content-map.md statuses.
3. 5 SPR-only items (answer entry practice).
4. Optional: Desmos 15-minute graphing drill.

Official resources

College Board (primary)

Resource	URL	Use
Math section overview	https://satsuite.collegeboard.org/sat/whats-on-the-test/math/overview	Domains + question counts
Digital SAT specs overview (PDF)	https://satsuite.collegeboard.org/media/pdf/digital-sat-test-spec-overview.pdf	Timing, adaptive design
Bluebook practice tests	https://satsuite.collegeboard.org/practice/practice-tests/bluebook	Full-length timed practice
Student Question Bank	Via Bluebook / My Practice after practice tests	Filter by domain, skill, difficulty
Digital SAT sample questions	Search College Board “Digital SAT sample questions” PDF	Public worked samples

Built-in tools

- **Desmos Graphing Calculator** inside Bluebook (practice outside: <https://www.desmos.com/calculator>)
- Personal approved calculator if preferred — still practice Desmos for test day

Free practice partners

- **Khan Academy Official SAT practice** (linked from College Board / My Practice tailored practice)

How we use these in this repo

- Lesson examples in weekend - */ are **instructor-created SAT-style** items unless marked otherwise.
- For authentic College Board items: pull from Question Bank / Bluebook, then log misses against skill IDs in 01-math-content-map.md.
- Do **not** paste full copyrighted Bluebook forms into this repo.

Preparation plan

This file is SAT prep only. It is intentionally separate from the Common App / supplementary-essay packet under docs/.

Do not merge SAT materials into docs/supplemental-essays-and-plans-2026-27.md.

Repo separation (important)

Workstream	Location	Purpose
Common App / essays / college list	docs/	Supplemental essay prompts, ED/EA/RD deadlines
SAT Math preparation	sat-prep/ (this folder)	Digital SAT structure, Math content map, weekend lesson plans

Executive Summary

Build an **up-to-date Digital SAT Math** teaching packet inside **sat -prep/** in the existing Karina_CommonApp repo, focused on raising Math from ~**680** → ~**780**.

Primary deliverables:

1. A short **Digital SAT structure primer** (what changed vs the old paper SAT).
2. A complete **College Board Math content map** (4 domains → concrete skills).
3. A **multi-weekend curriculum roadmap** aligned to your teaching model: *concepts first, then drill*.
4. **Fully concrete Weekend 1–4 lesson packs** for **Saturday 4h + Sunday 4h**, with: - minute-by-minute agenda - Q&A / nitpick mode (student already knows the basics) - Weekend 1 = **all four domains** at high level - Weekends 2–4 = deep dive on Advanced Math, PSDA, Geometry & Trig - practice sets in SAT style + answer keys - Desmos / Bluebook tips
5. Readable **Markdown** docs (optional PDF export). Not Jupyter notebooks.

No new GitHub repository. Everything lives under sat -prep/ in the current repo.

Student / coaching context (assumptions)

Item	Assumption	Needs confirm?
Practice Math score	~680	Exact Bluebook score report preferred
Total score	“1000 and 140” → likely 1140 (or 1040)	Confirm exact total + RW
Target Math	~780 (+100)	Confirm
Cadence	Sat 4h + Sun 4h	Fixed
Sequence	High-level survey of all four → then intensive practice / nitpick	Fixed
First focus	Math only (RW later)	Fixed unless you say otherwise

Important score math check: If Math is 680 and total is 1040, Reading & Writing would be ~360. If total is 1140, RW $\approx$ 460. Please confirm the score report so we don't mis-prioritize later.

What “current SAT Math” means (research basis)

We will ground materials in **College Board Digital SAT Suite** specs (not 2016–2023 paper SAT):

- **Platform:** Bluebook app; multistage **adaptive** (Module 2 harder/easier based on Module 1).
- **Math timing:** 2 modules $\times$ **35 minutes, 22 questions** each (20 scored + 2 pretest) $\rightarrow$ **44 Q / 70 min.**
- **Formats:** ~75% multiple choice, ~25% student-produced response (SPR).
- **Calculator:** Allowed **entire** Math section; built-in **Desmos** graphing calculator + personal approved calculator.
- **Ordering:** Within a module, generally easier $\rightarrow$ harder.
- **Domains (every module mixes all four):**

Domain	Approx. share	# questions	Core skills
Algebra	~35%	13–15	Linear equations/inequalities, linear functions, systems
Advanced Math	~35%	13–15	Nonlinear: quadratic, exponential, absolute value, polynomials, rationals, radicals
Problem-Solving & Data Analysis (PSDA)	~15%	5–7	Ratios/%, units, 1- and 2-var data, probability, stats claims
Geometry & Trigonometry	~15%	5–7	Area/volume, angles/triangles, right-triangle trig, circles

Sources to cite in docs: College Board Math Overview, Digital SAT Specs Overview PDF, Official Bluebook practice, Student Question Bank.

Authentic practice policy (copyright-safe)

We will **not** dump full Bluebook practice tests or pirated “real past papers” into the repo.

Instead:

1. **Link + cite** College Board public sample questions / Official Digital SAT sample PDFs.
2. Teach how to pull filtered items from the **Official Student Question Bank** (by domain/skill/difficulty).
3. Write **original SAT-style** questions matching each skill (with full solutions), clearly labeled as instructor-created.
4. Optional: paste only items College Board explicitly publishes for free educational use, with attribution and source URL.

This keeps the packet usable, legal, and still “exam-real.”

Repo layout (to create)

```
sat-prep/  
  README.md                # how instructor + student use  
this folder  
  CURRICULUM.md            # multi-weekend roadmap + goals  
  00-digital-sat-overview.md # structure, scoring,  
Bluebook/Desmos changes  
  01-math-content-map.md    # full skill checklist by domain  
  resources.md              # Bluebook, Question Bank, Khan,  
Desmos links  
  weekend-01/  
    README.md              # Sat/Sun overview + goals  
    saturday.md            # 4-hour script + concepts +  
examples + set  
  sunday.md                # 4-hour script + concepts +  
examples + set  
  practice-set-01.md        # mixed practice for weekend 1  
  answer-key-01.md          # full worked solutions  
  weekend-02/ ... weekend-04/ # full Sat/Sun packs (not stubs)  
  (optional later) pdf/     # exported PDFs if tooling  
available
```

Common App docs stay untouched under docs/.

Teaching design

Multi-weekend arc (Math-only)

Weekend	Theme	Goal
1	All four domains at high level (Q&A / nitpick) + mixed diagnostic	She already knows the basics; diagnose leaks
2	Advanced Math deep dive (equivalent expressions, abs/rational/radical, exponentials, nonlinear systems)	Accuracy on ~35% of test
3	PSDA (ratios, %, data, probability, margin of error, study claims)	Kill careless wording / table misses
4	Geometry & Trigonometry + Module 1 mixed simulation	Cover remaining ~15% + pacing
5	Full Bluebook Math section(s) + error log clinic	Convert concepts → score
6+	Targeted weak-skill cram + timed modules	Push toward 780

PSDA = Problem-Solving and Data Analysis. Not a fifth subject — it is College Board's name for the story / table / percent / probability slice.

Weekend 1 is a **tour of all four**. Weekends 2–4 go deep. 5+ = official timed practice.

Weekend 1 concrete agenda (updated)

Student is familiar with A–D. Saturday is **not** an Algebra-only course.

Saturday — 4 hours (all four domains)

Block	Time	Content
0	0:00–0:20	Orientation + name the four domains + what PSDA means

Block	Time	Content
1	0:20–1:10	Algebra A1–A5 Q&A + rapid items
2	1:10–2:00	Advanced Math B1–B4 Q&A + rapid items
Break	2:00–2:10	
3	2:10–3:00	PSDA C1–C7 Q&A + rapid items
4	3:00–3:50	Geometry & Trig D1–D4 Q&A + rapid items
5	3:50–4:00	Exit ticket: one item per domain

Sunday — 4 hours (mixed diagnostic + nitpick)

Block	Time	Content
0	0:00–0:20	Saturday miss review
1	0:20–0:45	Timed 12-item mixed diagnostic (all four)
2	0:45–1:40	Nitpick clinic on the diagnostic
Break	1:40–1:50	
3	1:50–3:10	Four-domain pressure stations (student talks setup first)
4	3:10–3:50	SPR + Desmos + Module 1 habits
5	3:50–4:00	16-item all-domain take-home

Each content block is **Q&A first**, then items, then a 5-minute trap recap — not a from-scratch mini-lesson. Extra linear reps live in `weekend-01/optional-algebra-drill.md` if Algebra is actually sloppy.

Execution phases

Phase 0 — Clarify scores (parallel, non-blocking)

Capture exact Math/RW/total + Bluebook practice test number if known. Defaults proceed with Math ~680 → ~780.

Acceptance: Assumptions logged in `sat-prep/README.md`.

Phase 1 — Foundations docs

Write:

- `00-digital-sat-overview.md`
- `01-math-content-map.md` (every College Board skill listed as a checklist)
- `CURRICULUM.md`
- `resources.md`
- `root sat-prep/README.md`
- Update repo root `README.md` to link Common App docs + SAT prep

Acceptance: Accurate Digital SAT facts; domain table matches College Board; no paper-SAT leftovers (no separate no-calc section).

Phase 2 — Weekend 1 full teaching pack (all four domains)

Write saturday/sunday scripts, practice set, answer key. Saturday covers A+B+C+D at high level.

Acceptance: Instructor can teach Weekend 1 from the markdown alone without inventing structure.

Phase 3 — Weekends 2–4 full packs

Full Sat/Sun scripts + practice sets + answer keys for Advanced Math, PSDA, Geometry & Trig.

Acceptance: Each weekend folder is teachable the same way as Weekend 1.

Phase 4 — Optional PDF

If pandoc/weasyprint available, export Weekend 1 pack to PDF; otherwise keep Markdown as source of truth.

Phase 5 — Commit + push

Commit on current branch (or a new `cursor/sat-math-prep-...` branch if cleaner) and push to the same GitHub repo.

Testing / QA

- Fact-check overview against College Board specs PDF + Math overview page.
 - Every skill in content map maps to a College Board domain skill name.
 - Practice items tagged by domain/skill/difficulty.
 - Answer key independently re-solved.
 - Copyright review: no unauthorized full-form dumps.
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Out of scope (unless you expand)

- Full Reading & Writing curriculum
 - Live interactive web app / Khan clone
 - Guaranteeing a specific score
 - Paying for third-party question banks or uploading copyrighted Bluebook forms
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What I need from you (optional but helpful)

1. Exact practice score report (Math / RW / total; Bluebook test # if any).
2. Next real SAT date (or target month).
3. After Weekend 1, which domain leaked most (use that to decide extra Question Bank filters).

4. Preference: Markdown only, or Markdown + PDF.

Execution status (updated 2026-08-20)

Phase	Status
0 Assumptions in README	Done
1 Foundations docs	Done
2 Weekend 1 full pack (all four domains, Q&A)	Done — see weekend-01/
3 Weekends 2–4 full packs	Done — see weekend-02/ ... weekend-04/
4 Optional PDF	Skipped (Markdown is source of truth)
5 Commit + push	On cursor/sat-weekend-packs-f685

Next (only if requested): Weekend 5 Bluebook clinic notes; PDF export; plug in exact scores / test date.

Weekend 1 overview

Goal: High-level Q&A coverage of **all four** Digital SAT Math domains, then a mixed diagnostic. Karina already knows the basics — Saturday is a tour + trap clinic, not a from-scratch Algebra course.

Time: Saturday 4h + Sunday 4h

Day	File	Focus
Saturday	saturday.md	Orientation + Algebra + Advanced Math + PSDA + Geometry (one block each)
Sunday	sunday.md	Mixed timed diagnostic + nitpick stations + SPR/Desmos habits
Practice	practice-set-01.md	16-item take-home, all four domains, 24 min
Answers	answer-key-01.md	Full worked solutions
Optional	optional-algebra-drill.md	Extra linear reps only if Algebra is sloppy

PSDA = Problem-Solving and Data Analysis (ratios, percents, data, probability, statistical claims). ~15% of Math.

Homework after Weekend 1: Official Question Bank — 8 items in the weakest domain + 4 in the strongest. Log misses on the skill map.

Weekend 1 Saturday

Theme: High-level Q&A survey of **all four** Digital SAT Math domains

Skills: A1–A5 · B1–B4 · C1–C7 · D1–D4 (overview, not deep teach)

Mode: Student talks first. Instructor nitpicks. Do **not** lecture from zero.

Materials: This doc, Desmos, scratch paper, timer, skill map `../01-math-content-map.md`

All practice items below are **instructor-created SAT-style** (not copied from Bluebook).

Karina already has a basic handle on these topics. Saturday is a **tour + trap clinic**: what SAT actually asks, where ~680 scorers leak points, and which skills need Weekend 2–4 depth.

If Algebra is sloppy and you want extra linear reps, use the optional bank: `optional-algebra-drill.md`.

Block 0 — Orientation + the four domains (0:00–0:20)

Talk track (short)

1. Math is **70 minutes, 2 modules × 35 minutes, 22 questions each**.
2. **Calculator + Desmos** on every question. Algebra fluency still wins time.
3. Module 1 accuracy routes Module 2 difficulty — do **not** gift easy misses.
4. ~25% are **SPR** (type the number). Blank = free miss.
5. Pacing ≈ **1.6 min/Q**. Stuck >2.5 min → flag and move.
6. Every module mixes **all four** domains. That is why today covers all four.

What the four domains are (say this out loud)

Tag	Name	Share	In one sentence
A	Algebra	~35%	Lines, linear equations, systems, inequalities
B	Advanced Math	~35%	Nonlinear: quadratic, exponential, abs value, rational, radical
C	PSDA	~15%	Problem-Solving and Data Analysis — ratios, %, data, probability, stats claims
D	Geometry & Trig	~15%	Area/volume, triangles, right-triangle trig, circles

PSDA is College Board’s name, not a new subject. It is the “story + table + percent + probability” slice. Only 5–7 questions, but they are easy points to drop if wording is sloppy.

Student do-now (3 min)

Name the four domains from memory. Then, without calculating yet, say **which domain** this is and **first move**:

A ball's height is $(h(t) = -16t^2 + 32t + 5)$. When does it hit the ground?

Expected: Domain B (quadratic). Set $(h=0)$ and solve / Desmos root. Do **not** grind the formula in the do-now.

How each content block runs (instructor rule)

For Blocks 1–4, do **not** teach a full mini-lesson first.

1. **Oral Q&A (student answers first)** — 8–10 min
2. **Rapid written items** — student works; you only interrupt on a trap — ~30 min
3. **Nitpick recap** — 5 min: “what almost got you”

If a block finishes early, add one harder oral: “What would SAT change to make this a hard item?”

Block 1 — Algebra survey (0:20–1:10) · A1–A5

Oral Q&A (student first)

1. You see $(5 - 2(3x + 1) = x - 17)$. What is the first trap, before any arithmetic?
2. When do two linear equations have **no** solution? When infinitely many?
3. When must you **flip** an inequality? Give an example.
4. In $(C = 12n + 45)$, what do 12 and 45 *mean*? What question is SAT likely to ask besides “find (C)”?
5. System in context: adult \$12, child \$7, 15 tickets, \$145. What are the two equations?

Rapid items

S1. (A1) $(5 - 2(3x + 1) = x - 17)$

S2. (A1, SPR) $\left(\frac{x}{4} + \frac{x}{6} = 5\right)$

S3. (A4) How many solutions? $\begin{cases} 2x-4y=6 \\ x-2y=5 \end{cases}$

S4. (A5) Solution set of $(-2x+5>13)$:

- A) $(x>-4)$ B) $(x<-4)$ C) $(x>4)$ D) $(x<4)$

S5. (A2–A3) Write $(3x-2y=12)$ in slope-intercept form. State slope and (y)-intercept.

S6. (A3, SPR) Gym: \$40 signup + \$25/month. Total paid \$165. How many months?

Nitpick recap

- Distributing a leading minus
 - LCD must hit **every** term
 - Parallel lines vs same line
 - Flip only when multiplying/dividing by a **negative**
 - Re-read: did they want (x), (y), or a *count*?
-

Block 2 — Advanced Math survey (1:10–2:00) • B1–B4

Oral Q&A (student first)

1. SAT says “which expression is **equivalent**.” What is your fastest check besides expanding everything?
2. How do you find the vertex of $(y=ax^2+bx+c)$ without completing the square?
3. Table: (y) goes 12, 18, 27, 40.5. Linear or exponential? How do you know in 15 seconds?
4. Why can a rational equation grow an **extraneous** root?
5. A line and a parabola: how many intersections are possible? How does Desmos settle it?

Rapid items

S7. (B1) Which is equivalent to $(4x(x-3)-2(x-3))$?

- A) $(2(x-3)(2x-1))$ B) $(2(x-3)(2x+1))$ C) $((x-3)(4x-2))$ D) Both A and C

S8. (B4) Vertex of $(y=-2(x+1)^2+5)$. Maximum or minimum?

S9. (B4) Which models exponential **decay**?

A) $(y=20-3x)$ B) $(y=20(0.85)^x)$ C) $(y=20(1.15)^x)$ D) $(y=3x+20)$

S10. (B2) Solve $(|2x-3|=7)$.

S11. (B3) How many solutions: $(y=x^2)$ and $(y=x+2)$?

S12. (B2, SPR) For $(y=x^2-4x-1)$, the minimum value of (y) is?

Nitpick recap

- Factor the common binomial before expanding
 - Vertex form: $((h,k))$ is opposite the sign inside $((x-h))$
 - Decay = multiply by a factor **between 0 and 1**, not “minus a constant”
 - Absolute value: two cases, then check
 - Nonlinear systems: algebra *or* Desmos — both, if time
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Break (2:00–2:10)

Stretch + water. No phones if possible.

Block 3 — PSDA survey (2:10–3:00) · C1–C7

Say **the name**: Problem-Solving and Data Analysis. Not “statistics class.” SAT’s version is short and wording-heavy.

Oral Q&A (student first)

1. “Increases by 20%” vs “increases by 20.” What’s the equation move?
2. A value goes **up 25% then down 25%**. Back to start? Why or why not?
3. When is **median** a better center than **mean**?
4. Two-way table: what’s the difference between $(P(\text{A}))$ and $(P(\text{A} \mid \text{B}))$?
5. A poll is $(52\% \pm 4\%)$. What *can* you conclude? What can you **not**?

Rapid items

S13. (C1) A mix uses flour:sugar ($=2:5$). If you use 12 cups of flour, how many cups of sugar?

S14. (C2) A price of \\$80 increases 25%, then the new price decreases 25%. What is the final price?

S15. (C3) Data: (2,2,3,7,16). Mean and median? Which is pulled by the outlier?

S16. (C4) A scatterplot of study hours vs score is roughly linear and rising. The line of best fit is ($y=8x+42$). Interpret the slope. Predict the score at 6 hours.

S17. (C5) Club members:

	Yes	No	Total
Junior	12	8	20
Senior	18	12	30
Total	30	20	50

What is the probability a **randomly chosen senior** said Yes?

S18. (C6–C7) A random sample of voters gives 52% support, margin of error 4%. Which conclusion is most appropriate?

- A) Exactly 52% of all voters support the candidate.
- B) The true support is likely between 48% and 56%.
- C) The candidate will win.
- D) The sample was biased because it is not 50–50.

Nitpick recap

- Percent of **what** (original vs new)
 - Successive percents do **not** cancel
 - Mean follows extremes; median often doesn't
 - Conditional probability: restrict the row/column first
 - Margin of error $\neq$ "the answer is exactly the sample percent"
-

Block 4 — Geometry & Trig survey (3:00–3:50) · D1–D4

Oral Q&A (student first)

1. If every length is scaled by 2, what happens to **area**? To **volume**?
2. Recite 45-45-90 and 30-60-90 side recipes.
3. $(\sin\theta=)$? $(\cos\theta=)$? $(\tan\theta=)$?
4. Similar triangles: what stays the same, what scales?
5. Circle radius 6, central angle 60° . How do you get arc length without memorizing a random formula?

Rapid items

S19. (D1) A cube has edge 3. Volume? If the edge is doubled, new volume?

S20. (D2) Two similar triangles. A pair of corresponding sides is 6 and 8. A side of 9 on the smaller triangle corresponds to what on the larger?

S21. (D3) Isosceles right triangle, leg 5. Hypotenuse?

S22. (D3) Right triangle, opposite 3, hypotenuse 5. $(\sin\theta=)$?

S23. (D4) Circle radius 6. Arc measure 60° . Arc length?

S24. (D1–D3, SPR) Rectangle 8 by 6. Length of a diagonal?

Nitpick recap

- Linear scale (k) → area (k^2), volume (k^3)
 - Similarity is a **ratio**, not “add 2 to each side”
 - Special rights beat Pythagorean when you recognize them
 - Trig is a **ratio** — cancel units, don’t invent extra lengths
 - Arc length = fraction of circumference $(\theta/360) \cdot 2\pi r$
-

Block 5 — Exit ticket + homework (3:50–4:00)

Exit ticket (4 items, one per domain, timed 5 minutes)

- E1. (A) $4(x-1)-x=2x+5$
E2. (B) Zeros of $(f(x)=2(x-3)(x+1))$
E3. (C) 40 is what percent of 160?
E4. (D) Area of a circle with radius 4 (in terms of (π))

Assign

1. Mark `../01-math-content-map.md`:  if rusty,  if automatic,  if a miss today.
 2. Before Sunday: Official Question Bank, **5 items from each domain** (20 total), mixed difficulty.
 3. Bring a list of “I hesitated” items — those are Sunday’s first 20 minutes.
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Instructor notes

- Today’s win is **coverage + diagnosis**, not mastery of any one skill.
- If Algebra is already automatic, spend leftover Block 1 minutes on S3-type “how many solutions” items — those are Module 1 gifts.
- If PSDA wording is the slow leak, do not “save it for Weekend 3.” Flag it, and Sunday will hit it again.
- Track every miss: skill ID + type (concept / setup / arithmetic / rushed / wording).

Weekend 1 Sunday

Theme: Mixed diagnostic + nitpick clinic across **all four** domains

Skills: A · B · C (PSDA) · D

Mode: Timed reps, then student-explains-the-miss. Still not a from-scratch lecture.

Block 0 — Saturday miss review (0:00–0:20)

Re-work every Saturday miss **cold** (no notes). If still wrong → fix the micro-gap immediately.

Fast-fail checklist (whiteboard)

1. Did I distribute the negative?
 2. Did I flip the inequality?
 3. Did I check the solution in the **original** equation?
 4. For systems: did I solve for **both** variables?
 5. Percent of *what*? Conditional probability: which row?
 6. Scale factor: length vs area vs volume?
 7. SPR: sign, fraction, decimal — what Bluebook will accept?
-

Block 1 — Timed mixed diagnostic (0:20–0:45) · all four domains

12 questions, 18 minutes. Calculator/Desmos allowed. Student works silently. You only watch pacing.

D1. (A1) $7-3(2x-1)=2x+8$

D2. (A4) Solution of $\begin{cases} x+y=10 \\ 2x-y=5 \end{cases}$

D3. (A5) $(-\frac{12}{x+3} \leq 7)$. Solution set?

D4. (B1) Factor $(x^2-5x-14)$. Enter the **smaller** constant term of the two binomials (SPR).

D5. (B2) (x)-coordinate of the vertex of $(y=x^2-8x+10)$

D6. (B4) Population starts at 200 and grows 50% each decade. Population after 2 decades? (SPR)

D7. (C1) A car goes 180 miles in 3 hours. At that rate, miles in 2.5 hours? (SPR)

D8. (C2) A \$60 jacket is discounted 15%, then tax of 8% is added to the sale price. Final price?

D9. (C5) From Saturday’s club table (20 juniors, 30 seniors; 30 Yes, 20 No). ($P(\text{Yes})$) for a random member?

D10. (D3) Legs 5 and 12. Hypotenuse?

D11. (D2) A triangle has angles (40°) and (65°) . Third angle?

D12. (D4) Circumference of a circle with diameter 10, in terms of (π) .

Block 2 — Nitpick clinic on the diagnostic (0:45–1:40)

Go item by item **only where she missed or hesitated**. Student explains first.

Instructor prompts (use these; don’t redo a lecture)

- “Point to the first decision you made.”
- “Which domain is this, and why does that change the tool?”
- “If SAT swapped one word, where would you break?”
- “Desmos or hand? Why?”

Likely 680-band leaks (check even if she got them right)

Item	Typical leak
D1	Minus times a grouped term
D3	Forgot to flip
D6	Used $(+50)$ instead of $(\times 1.5)$
D8	Taxed the original $\$60$ instead of the sale price
D9	Used a row (conditional) instead of the whole table
D10	$(5+12=17)$ instead of Pythagorean

If the diagnostic is mostly clean, spend leftover time on **harder oral variants** of D6, D8, D9.

Break (1:40–1:50)

Block 3 — Four-domain pressure stations (1:50–3:10)

Three items per domain. Student talks the setup **before** calculating. You nitpick the setup.

Station A — Algebra

U1. Phone plan \$35/month + \$0.10 per text. Total \$47. How many texts?

U2. Which line is perpendicular to $(y=3x-1)$ and passes through $((0,4))$?

A) $(y=3x+4)$ B) $(y=-\frac{1}{3}x+4)$ C) $(y=\frac{1}{3}x+4)$ D) $(y=-3x+4)$

U3. Adult tickets \$12, child \$7. 15 tickets cost \$145. How many adult tickets? (SPR)

Station B — Advanced Math

U4. Expand $((x-4)^2)$.

U5. A ball's height $(h(t)=-16t^2+32t+5)$ (feet, seconds). When does it hit the ground? (exact form OK)

U6. Table $(x=0,1,2,3)$ and $(y=12,18,27,40.5)$. Linear or exponential? Write a model.

Station C — PSDA

U7. 18 is 45% of what number?

U8. Scores: 70, 74, 74, 76, 92. Mean vs median. Which should you quote if you want the “typical” score without the 92?

U9. A study asks only students in the honors hallway whether they like the new schedule, then claims “the school supports it.” What is the design problem? (C7)

Station D — Geometry & Trig

U10. Similar triangles. Corresponding sides 4 and 10. Smaller perimeter 18. Larger perimeter?

U11. 30-60-90 triangle, short leg 6. Long leg and hypotenuse?

U12. Right triangle, adjacent 8, hypotenuse 10. $(\cos\theta=)$?

Block 4 — SPR + Desmos + Module 1 habits (3:10–3:50)

SPR drill (8 minutes, answers only)

P1. $0.2(x+15)=0.5x-3$

P2. Water tank ($W=120-4t$). Minutes until empty?

P3. Bacteria double every 3 hours from 100. How many after 9 hours?

P4. Rectangle 5 by 12. Diagonal?

P5. (20\%) of 45?

Desmos drill (10 minutes)

1. Graph ($y=3(x-2)$) and ($y=2x+5$); read the intersection.
2. Graph ($y=x^2$) and ($y=x+2$); confirm Saturday S11.
3. Graph ($y=|2x-3|$) and ($y=7$); confirm S10.

Module 1 talk (last 12 minutes) — oral only

1. Why does missing two “easy” Algebra items on Module 1 hurt a 780 more than missing one hard Advanced Math item?
 2. When is Desmos faster than algebra? When is it slower?
 3. What do you do at 2:30 stuck?
 4. Name one PSDA wording trap and one Geometry scale-factor trap from today.
-

Block 5 — Recap + take-home (3:50–4:00)

Recap (oral)

1. What does **PSDA** stand for, and why is it only ~15% but still worth today’s time?
2. Name the four domains.
3. Why does Module 1 accuracy matter?
4. Which domain felt least automatic? That is the first target for Weekends 2–4.

Assign

1. Complete practice-set-01.md — **16 mixed items, all four domains**, timed **24 minutes**.
 2. Grade with answer-key-01.md; fill the error log.
 3. Question Bank: 8 items in the weakest domain + 4 in the strongest (keep the strong one sharp).
 4. Preview: Weekend 2 goes **deep** on Advanced Math (B1–B4). Weekend 3 = PSDA. Weekend 4 = Geometry & Trig.
-

Instructor notes

- Sunday is a **pressure test of Saturday's survey**, not a second survey.
- If one domain is a disaster, do not rebuild a full lesson. Isolate 1–2 skill IDs and write them on the content map as .
- Target feeling by end of Sunday: “I know what all four domains look like on sight, and I know where I leak.”

Weekend 1 practice set

Instructions: 16 questions. Target time **24 minutes**. Calculator/Desmos allowed.

Items are **SAT-style originals** covering Algebra, Advanced Math, PSDA, and Geometry & Trig.

Answers: answer-key-01.md.

Skill tags: A Algebra · B Advanced Math · C PSDA · D Geometry & Trig

1. **(A1, MC)** What is the solution of $7-3(2x-1)=2x+8$?
A) $x=\frac{1}{4}$ B) $x=0$ C) $x=1$ D) $x=2$
2. **(A1, SPR)** Solve $\frac{5x-2}{4}=\frac{x+6}{2}$.
3. **(A2, MC)** The line $5x+2y=10$ has slope
A) $-\frac{5}{2}$ B) $-\frac{1}{5}$ C) $\frac{1}{5}$ D) $\frac{5}{2}$

4. **(A4, MC)** Solution to $\begin{cases} x+y=10 \\ 2x-y=5 \end{cases}$ is
 A) $((5,5))$ B) $((4,6))$ C) $((6,4))$ D) $((3,7))$
 5. **(B1, MC)** Which expression is equivalent to $(2x(x+4)-(x+4))$?
 A) $((x+4)(2x-1))$ B) $((x+4)(2x+1))$ C) $(2x^2+7x-4)$ D) Both A and C
 6. **(B2, MC)** Zeros of $(f(x)=2(x-3)(x+1))$ are
 A) (3) and (-1) B) (-3) and (1) C) (2) and (0) D) (6) and (-2)
 7. **(B2, SPR)** For $(y=x^2-8x+10)$, the (x)-coordinate of the vertex is?
 8. **(B4, MC)** Which represents exponential growth?
 A) $(y=9-0.2x)$ B) $(y=9(0.8)^x)$ C) $(y=9(1.2)^x)$ D) $(y=\frac{9}{x})$
 9. **(C1, SPR)** A recipe uses oil:vinegar (=3:1). If you use 2 cups of vinegar, how many cups of oil?
 10. **(C2, MC)** A price of $\$80$ increases 25%, then decreases 25%. Final price?
 A) $\$80$ B) $\$75$ C) $\$85$ D) $\$70$
 11. **(C3, MC)** Data: (2,2,3,7,16). The median is
 A) 2 B) 3 C) 6 D) 7
 12. **(C5, MC)** Using the club table (Juniors 12 Yes / 8 No; Seniors 18 Yes / 12 No), $(P(\text{Yes} \mid \text{Senior}))$ is
 A) $(18/50)$ B) $(18/30)$ C) $(30/50)$ D) $(18/20)$
 13. **(D1, SPR)** Volume of a cube with edge 3.
 14. **(D2, MC)** A triangle has angles (40°) and (65°) . The third angle is
 A) (65°) B) (75°) C) (85°) D) (95°)
 15. **(D3, MC)** Isosceles right triangle with leg 5. Hypotenuse?
 A) 5 B) $(5\sqrt{2})$ C) $(5\sqrt{3})$ D) 10
 16. **(D4, SPR)** Circumference of a circle with diameter 10. Enter the number that multiplies (π) (so if the answer is (10π) , enter 10).
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After scoring

Missed #	Skill ID	Mistake type (concept / setup / arithmetic / rushed / wording)	Fix

Weekend 1 optional Algebra drill

Use **only if** Saturday Block 1 showed sloppy linear fluency and you want extra reps.

These are the original Weekend 1 Algebra items (instructor-created). Answers are in answer - key - 01.md § Optional Algebra drill.

A1

- $(5-2(3x+1)=x-17)$
- $(\frac{x}{4}+\frac{x}{6}=5)$
- (SPR) $(0.2(x+15)=0.5x-3)$
- A number decreased by 18 is four times the number. Find the number.
- $(\frac{3(2x-1)}{5}=x+1)$

A2–A3

- Write $(3x-2y=12)$ in slope-intercept form; state slope and (y)-intercept.
- Line with slope $(-\frac{1}{2})$ through $((-4,7))$. Equation in point-slope then slope-intercept.
- Cost $(C=12n+45)$ for (n) shirts. Interpret slope and intercept. What is (C) when $(n=8)$?
- Which line is perpendicular to $(y=3x-1)$ and passes through $((0,4))$?
A) $(y=3x+4)$ B) $(y=-\frac{1}{3}x+4)$ C) $(y=\frac{1}{3}x+4)$ D) $(y=-3x+4)$

A4

- $(\begin{cases} y=2x-3 \\ 3x+2y=26 \end{cases})$
- $(\begin{cases} 4x+3y=10 \\ 2x-y=1 \end{cases})$
- How many solutions? $(\begin{cases} 2x-4y=6 \\ x-2y=5 \end{cases})$
- (SPR) Adult tickets \$12, child \$7. 15 tickets cost \$145. How many adult tickets?

A5

14. Solve: $2(x-4) < 5x+1$.

15. Which point satisfies $y > 2x-1$ and $x+y \leq 4$?

A) $((0,5))$ B) $((1,2))$ C) $((3,0))$ D) $((2,3))$

16. For which (k) is $(x=2)$ a solution of $(3x+k > 10)$?

Weekend 2 overview

Goal: Turn “I know quadratics/exponentials” into automatic B1–B4, including abs/rational/radical traps.

Time: Saturday 4h + Sunday 4h

Day	File	Focus
Saturday	saturday.md	Equivalent expressions, absolute value, rationals, radicals
Sunday	sunday.md	Exponentials, nonlinear systems, function reading, timed B mix
Practice	practice-set-02.md	16-item take-home, 24 min
Answers	answer-key-02.md	Worked solutions

Homework: Question Bank → Advanced Math, medium/hard, 15–20 items. Log B1–B4 on the skill map.

Weekend 2 Saturday

Theme: Advanced Math deep dive (already-known concepts → SAT traps)

Skills: B1 equivalent expressions · B2 abs/rational/radical · B4 parameters

Mode: Student talks first. Instructor nitpicks. Short reminder, then volume.

All items are **instructor-created SAT-style**.

Weekend 1 already surveyed B. Today is **harder B1/B2**: the items that separate ~680 from ~750.

Block 0 — Target list (0:00–0:15)

From Weekend 1, circle which of these leaked:

- Equivalent expressions (factor vs expand)
- Absolute value cases
- Rational equations / **extraneous** roots
- Radical equations / **extraneous** roots
- Exponent rewrite

Today's rule: if she can say the *first move* correctly, let her finish. Interrupt only on a trap.

Block 1 — Equivalent expressions, harder (0:15–1:15) · B1

Oral (student first)

1. Fastest SAT check for “which is equivalent”: expand all, factor the stem, or Desmos overlay? When each?
2. Why must you state $(x \neq 3)$ when canceling $((x-3))$? Does SAT still call the simplified form equivalent?
3. Difference of cubes: $(a^3-b^3=)$?

Reminder (2 min, then items)

Factor / expand / cancel common factors / exponent rules. Desmos: identical graphs $\Rightarrow$ equivalent (watch domain holes).

Items

S1. Which is equivalent to $(\frac{x^2-9}{x-3})$ for $(x \neq 3)$?

- A) $(x-3)$ B) $(x+3)$ C) (x^2-3) D) $(3-x)$

S2. Simplify $(\frac{2x^3-8x}{2x})$ for $(x \neq 0)$.

S3. Factor completely: $(8x^3-27)$.

S4. Which is equivalent to $(4x(x-3)-2(x-3))$?

- A) $(2(x-3)(2x-1))$ B) $(2(x-3)(2x+1))$ C) $(4x^2-10x+6)$ D) $(2(x-3)(2x-1)+1)$

S5. (SPR) If $((3x^2 y)^2 = ax^b y^c)$ and $(a,b,c>0)$, what is $(a+b+c)$?

S6. Rewrite $(\sqrt{18x^2})$ as a simplified radical. Assume $(x>0)$.

Nitpick

- Cancel **factors**, not terms ((x^2-9) is not (x^2) over 9)
 - Cubes: $(8x^3-27=(2x-3)(4x^2+6x+9))$ — the quadratic does **not** factor over the reals
 - $((3x^2 y)^2=9x^4 y^2)$, not $(3x^4 y^2)$
-

Block 2 — Absolute value (1:15–2:05) · B2

Oral

1. $(|A|=k)$ with $(k>0)$ means how many cases? If $(k=0)$? If $(k<0)$?
2. $(|A|<3)$ as a compound inequality?
3. $(|A|=|B|)$ — what's the clean case split?

Items

S7. Solve $(|3x+1|=10)$.

S8. Solve $(|x-4|<3)$. Write as a compound inequality.

S9. Solve $(|2x-5|=|x+1|)$.

S10. How many solutions: $(|x-2|=-5)$?

A) 0 B) 1 C) 2 D) infinitely many

S11. (SPR) The larger solution of $(|2x-3|=7)$.

Nitpick

- Absolute value is never negative — S10 is a freebie if she remembers that
 - After solving cases, **check** (especially with variables on both sides)
-

Break (2:05–2:15)

Block 3 — Rational equations (2:15–3:10) · B2

Oral

1. First move on a rational equation?
2. What value is never allowed?
3. You solve and get the forbidden value. What do you write?

Reminder

Multiply through by the LCD. Discard any solution that makes a denominator 0.

Items

S12. Solve $\frac{x+3}{x-1}=2$.

S13. Solve $\frac{1}{x-2}+3=\frac{5}{x-2}$.

S14. Solve $\frac{x}{x-1}=\frac{2}{x-1}+1$. How many solutions?

S15. For which (x) is $\frac{x^2-5x+6}{x-2}$ undefined? What is the simplified expression for all other (x)?

S16. (SPR) Solution of $\frac{4}{x}+\frac{1}{2}=3$. (Assume one real solution.)

Nitpick

- S14 is a **no-solution** trap: algebra can look like it “cancels” into a contradiction
 - Undefined at the hole vs removable discontinuity — SAT usually asks “undefined” or “equivalent for $(x \neq)$ ”
-

Block 4 — Radical equations (3:10–3:50) • B2

Oral

1. Why square both sides? What extra root can appear?
2. After solving, what **must** you do?
3. ($\sqrt{\text{something}}$) is never negative. How does that kill a candidate?

Items

S17. Solve ($\sqrt{2x+3}=x$). Check every candidate.

S18. Solve ($\sqrt{x+12}=x$).

S19. Solve ($\sqrt{3x-2}=4$).

S20. (SPR) The **larger** valid solution of ($\sqrt{5x+1}=x+1$). If none, enter 0.

Nitpick

- Isolate the radical **before** squaring
 - Check is not optional — this is the SAT skill
-

Block 5 — Exit + homework (3:50–4:00)

E1. Equivalent to ($\frac{x^2-16}{x-4}$) for ($x \neq 4$)?

E2. Solve ($|x+5|=9$).

E3. Does ($\sqrt{x}=x-2$) have solution ($x=1$)? Yes or no — and why?

Assign

Question Bank: Advanced Math, **equivalent expressions + nonlinear equations**, medium/hard, 20 items.

Bring Sunday: list of any rational/radical miss.

Weekend 2 Sunday

Theme: Exponentials, nonlinear systems, function reading, timed B mix

Skills: B2 · B3 · B4

Block 0 — Saturday miss review (0:00–0:20)

Re-work every miss cold. Extra reps if needed: one abs, one rational, one radical.

Whiteboard:

1. Equivalent $\neq$ “looks similar after dropping a term.”
 2. $(|A|=k<0) \rightarrow$ no solution.
 3. Forbidden denominator $\rightarrow$ discard.
 4. Radical solutions must be checked.
-

Block 1 — Exponential equations & models (0:20–1:15) · B2 / B4

Oral

1. “Grows by 6% per year” $\rightarrow$ what multiplier?
2. Same-base move: $(2^{x+1}=16)$ — rewrite 16 first.
3. Linear vs exponential from a table: differences vs ratios.

Items

U1. Solve $(2^{x+1}=16)$.

U2. Solve $(5 \cdot 3^x=45)$.

U3. $(P=100\left(\frac{12}{100}\right)^t)$. For which (t) is $(P=25)$?

U4. Which models exponential growth?

A) $(y=9-0.2x)$ B) $(y=9(0.8)^x)$ C) $(y=9(1.2)^x)$ D) $(y=9/x)$

U5. (SPR) Bacteria start at 80 and triple every 4 hours. How many after 12 hours?

U6. A car worth \$20,000 depreciates 15% per year. Write a model $(V(t))$. Find $(V(2))$.

Nitpick

- $(+6\%)$ is $(\times 1.06)$, not $(+6)$
 - Half-life / doubling: count **how many periods**, not clock hours blindly
-

Block 2 — Nonlinear systems (1:15–2:05) • B3

Oral

1. Line + parabola: 0, 1, or 2 intersections. How do you see it in Desmos?
2. Substitution vs graphing: when each?
3. What does a repeated root mean geometrically?

Items

U7. Solve $\begin{cases} y=x^2 \\ y=2x+3 \end{cases}$

U8. How many solutions: $\begin{cases} y=x^2+1 \\ y=x \end{cases}$?

A) 0 B) 1 C) 2 D) 3

U9. Solve $\begin{cases} x+y=5 \\ y=x^2-1 \end{cases}$

U10. (SPR) The **larger** (x) -value in the intersection of $(y=x^2-4)$ and $(y=x)$.

U11. A circle $(x^2+y^2=25)$ and line $(y=3)$. How many intersection points? Find them.

Nitpick

- After substituting, you still have to **pair** (y) with each (x)

- U8: ($x^2-x+1=0$) discriminant ($1-4<0$) → none — a graph would show the line missing the parabola
-

Break (2:05–2:15)

Block 3 — Reading nonlinear functions (2:15–3:05) • B4

Oral

1. $f(x)=a(x-h)^2+k$: what do (a,h,k) do?
2. $(g(x)=A \cdot r^x)$: what is $(g(0))$? What is (r) ?
3. SAT asks “which statement is true about the function” — they want a **parameter meaning**, not a new calculation.

Items

U12. $f(x)=2(x-1)^2+5$. Vertex? Min or max? $f(3)$?

U13. $g(x)=3 \cdot 2^x$. Find $(g(0))$ and $(g(4))$. Interpret 3 and 2.

U14. A ball $(h(t)=-16t^2+48t+4)$. What is the **maximum** height? (Desmos or vertex.)

U15. Which function has a (y) -intercept of 7 and decreases by a constant factor?

A) $(y=7-2x)$ B) $(y=7(0.6)^x)$ C) $(y=7+0.6x)$ D) $(y=0.6x+7)$

U16. (SPR) For $(y=-(x+2)^2+9)$, the maximum value of (y) .

Block 4 — Timed Advanced Math mix (3:05–3:50)

12 questions, 18 minutes (half-module density, B-heavy). Then 10 minutes of miss review.

- T1. Zeros of $(x^2+x-12=0)$.
- T2. Equivalent to $((x-4)^2)$?
- T3. $(\frac{6x^2-9x}{3x})$ for $(x \neq 0)$.
- T4. $(|x+5|=9)$. Solutions.
- T5. $(\frac{x+3}{x-1}=2)$. Solution.
- T6. $(\sqrt{3x-2}=4)$. Solution.
- T7. $(3^x=81)$.
- T8. Intersections of $(y=x^2)$ and $(y=2x+3)$: how many?
- T9. Vertex of $(y=-2(x+1)^2+5)$.
- T10. $(200(1.5)^2)$.
- T11. (SPR) Minimum of $(y=x^2-4x-1)$.
- T12. Which is exponential decay: $(y=20-3x)$ or $(y=20(0.85)^x)$?
-

Block 5 — Recap + take-home (3:50–4:00)

1. Complete `practice-set-02.md`, 16 items, **24 minutes**.
 2. Question Bank: 15 Advanced Math hard items.
 3. Preview Weekend 3: **PSDA** (Problem-Solving and Data Analysis) — ratios, %, data, probability, claims.
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Instructor notes

- If Saturday rationals/radicals were clean, steal 15 minutes from Block 0 for an extra nonlinear-system Desmos race.
- 780 requires B **automatic**. A slow but correct quadratic is still a Module 1 time leak.

Weekend 2 practice set

16 questions, 24 minutes. Desmos allowed. Answers: `answer-key-02.md`.

1. **(B1, MC)** $(\frac{x^2-9}{x-3})$ for $(x \neq 3)$ is equivalent to
A) $(x-3)$ B) $(x+3)$ C) (x^2-3) D) $(9-x)$
2. **(B1, SPR)** If $((3x^2 y)^2 = ax^b y^c)$ with $(a,b,c > 0)$, enter $(a+b+c)$.
3. **(B1, MC)** $(8x^3-27=)$
A) $((2x-3)^3)$ B) $((2x-3)(4x^2+6x+9))$ C) $((2x-3)(4x^2-6x+9))$ D) $((8x-27)(x^2+1))$
4. **(B2, MC)** Solutions of $(|3x+1|=10)$
A) $(3,-3)$ B) $(3,-\frac{11}{3})$ C) $(\frac{11}{3},-3)$ D) (3) only
5. **(B2, MC)** $(|x-4| < 3)$ is
A) $(x < 7)$ B) $(1 < x < 7)$ C) $(x < 1)$ or $(x > 7)$ D) $(x > 1)$
6. **(B2, SPR)** Larger solution of $(|2x-5|=|x+1|)$.
7. **(B2, MC)** $(|x-2|=-5)$ has how many solutions?
A) 0 B) 1 C) 2 D) infinitely many
8. **(B2, SPR)** Solve $(\frac{x+3}{x-1}=2)$. Enter (x) .
9. **(B2, MC)** $(\frac{x}{x-1}=\frac{2}{x-1}+1)$ has
A) $(x=1)$ B) $(x=2)$ C) no solution D) every $(x \neq 1)$
10. **(B2, MC)** Valid solution of $(\sqrt{2x+3}=x)$
A) (-1) only B) (3) only C) (-1) and (3) D) none
11. **(B2, SPR)** Solve $(\sqrt{3x-2}=4)$.
12. **(B4, MC)** $(2^{x+1}=16)$
A) $(x=2)$ B) $(x=3)$ C) $(x=4)$ D) $(x=8)$
13. **(B4, SPR)** Bacteria start at 80 and triple every 4 hours. After 12 hours?
14. **(B3, MC)** $(\begin{cases} y=x^2 \\ y=2x+3 \end{cases})$ has how many solutions?
A) 0 B) 1 C) 2 D) 3
15. **(B4, MC)** $(f(x)=2(x-1)^2+5)$ has
A) max 5 at $(x=1)$ B) min 5 at $(x=1)$ C) min 5 at $(x=-1)$ D) min 2 at $(x=1)$
16. **(B4, SPR)** Maximum value of $(y=-(x+2)^2+9)$.

Weekend 3 overview

PSDA is College Board’s name for ratios, rates, percents, data displays, probability, and statistical claims. About **15%** of SAT Math (5–7 questions).

Goal: Stop leaking “easy” wording points. Karina already knows the school-math; today is translation + tables + what you may conclude.

Time: Saturday 4h + Sunday 4h

Day	File	Focus
Saturday	saturday.md	C1 ratios/rates/units · C2 percents · C3 one-var data
Sunday	sunday.md	C4 scatterplots · C5 probability · C6–C7 MOE / study design
Practice	practice-set-03.md	16-item take-home, 24 min
Answers	answer-key-03.md	Worked solutions

Homework: Question Bank → PSDA, all skills, 20 items (include some C6/C7).

Weekend 3 Saturday

Theme: PSDA — Problem-Solving and Data Analysis

Skills: C1 ratios/rates/units · C2 percents · C3 one-variable data

Mode: Wording first. These are not “new math.” They are SAT’s favorite way to hide easy arithmetic.

What PSDA is: College Board’s name for the ~15% of Math (5–7 questions) that is ratios, unit rates, percents, tables/graphs, probability, and statistical *claims*. A 680 scorer often “knows” this and still drops 2–3 raw points on wording.

All items are **instructor-created SAT-style**.

Block 0 — PSDA in one page (0:00–0:15)

Oral

1. Say “PSDA” in words.
2. Why can a 5-question slice still move a score from 730 toward 780?
3. First move on every PSDA item: **underline the quantity they want.**

Whiteboard translation habits:

English	Math move
per, each, rate	divide; units in the answer
of	multiply
increases by ($p\%$)	$(\times(1+p/100))$
decreases by ($p\%$)	$(\times(1-p/100))$
what percent of	part/whole $(\times 100)$
given the mean of (n) values	sum $(= n \cdot \text{mean})$

Block 1 — Ratios, rates, units (0:15–1:20) • C1

Oral

1. (2:5) with 12 on the first part — how do you set the proportion?
2. 45 miles in $\frac{3}{4}$ hour — mph?
3. Two workers at different rates. Combined rate is **sum**, then time (=) work / combined.

Items

S1. Mix is flour:sugar (=2:5). 12 cups flour. Cups of sugar?

S2. A car travels 180 miles in 3 hours. At that rate, miles in 2 hours 30 minutes? (SPR)

- S3. Printer A: 40 pages/min. Printer B: 55 pages/min. Together, minutes to print 600 pages?
- S4. A recipe for 8 servings uses 3 cups of broth. Cups of broth for 20 servings?
- S5. Convert 90 minutes to hours, then find distance at 48 mph. (SPR, miles)
- S6. Juice concentrate:water (=1:3). How many cups of concentrate in 16 cups of finished mix?
- S7. (MC) Which rate is fastest?
- A) 5 miles in 12 minutes B) 20 mph C) 8 miles in 20 minutes D) 0.4 miles/minute

Nitpick

- Hours vs minutes (2.5 not 2:30 as 230)
 - Combined rate: add rates, **not** add times
 - Ratio parts: total parts first when the question gives the *whole*
-

Block 2 — Percentages (1:20–2:15) • C2

Oral

1. “40 is what percent of 160?” vs “40% of 160 is?”
2. Up 25% then down 25% — back to start?
3. Tax after discount: which price gets the tax?

Items

- S8. 40 is what percent of 160?
- S9. 18 is 45% of what number? (SPR)
- S10. \$80 increases 25%, then decreases 25%. Final price?
- S11. A \$60 jacket is 15% off, then 8% tax on the **sale** price. Final price?
- S12. A value increases from 50 to 65. Percent increase?

S13. A population is 800. It grows 10%, then grows 10% again. New population?

A) 960 B) 968 C) 880 D) 1000

S14. (SPR) What number, increased by 20%, becomes 72?

Nitpick

- Percent change ($=\frac{\text{new}-\text{old}}{\text{old}}$)
 - Successive 10% and 10% is ($\times 1.1 \times 1.1 = 1.21$), not (+20%)
 - “Increased by 20% becomes 72” $\rightarrow (1.2x=72)$, not $(x+20=72)$
-

Break (2:15–2:25)

Block 3 — One-variable data (2:25–3:35) · C3

Oral

1. Mean vs median vs range — one sentence each.
2. Add a huge outlier. What happens to each?
3. Mean of 5 scores is 80. What’s the sum? Why does SAT love that?

Items

S15. Data: (2,2,3,7,16). Mean, median, range?

S16. Scores: (70,74,74,76,92). Mean vs median. Which is the better “typical” if you want to ignore the 92?

S17. Five tests, mean 80. The first four scores are 72, 88, 90, 70. What is the fifth? (SPR)

S18. List A: (3,5,5,7,10). List B: (3,5,5,7,30). Which statement is true?

A) Medians differ a lot B) Means differ a lot C) Ranges are equal D) Both means equal 6

S19. Adding 100 to the set (2,2,3,7,16) will

- A) raise the median more than the mean
- B) raise the mean more than the median
- C) leave both unchanged
- D) lower the range

S20. A histogram is piled on the left with a long tail to the right (right-skewed). Typically:

- A) mean ($<$) median B) mean ($>$) median C) mean ($=$) median always D) range is 0

Nitpick

- Odd count: median is the middle after **sorting**
 - Range ($=$) max – min, not “how many numbers”
 - Skew: tail pulls the **mean**
-

Block 4 — Mixed C1–C3 sprint (3:35–3:55)

M1. $3:4 = (x:20)$. ($x=?$)?

M2. 12 is 15% of what?

M3. Mean of (4,6,10, x) is 8. ($x=?$)?

M4. 30% off \\$40, then extra \\$5 coupon. Final?

M5. 2.4 hours at 55 mph. Distance?

Block 5 — Exit + homework (3:55–4:00)

E1. Up 20% from 50, then down 20%. Result?

E2. Ratio 5:2, total 28. Larger part?

E3. Median of (9,3,3,11,7)?

Assign

Question Bank: PSDA → ratios, percents, one-variable data, 15 items.

Sunday is tables, probability, and “what can you conclude.”

Weekend 3 Sunday

Theme: Two-variable data, probability, inference, study-design claims

Skills: C4 · C5 · C6 · C7

Block 0 — Saturday miss review (0:00–0:15)

Re-work every C1–C3 miss. If percents were the leak, redo S10, S13, S14 cold.

Block 1 — Scatterplots and models (0:15–1:10) · C4

Oral

1. Positive vs negative association.
2. Line of best fit ($y=mx+b$): meaning of (m) and (b) in context.
3. Linear vs exponential: constant difference vs constant ratio.
4. A prediction *outside* the data range (extrapolation) — trust it?

Items

U1. Best fit ($y=8x+42$) for study hours (x) and score (y). Interpret slope. Predict ($x=6$).

U2. Table: ($x=0,1,2,3$) and ($y=12,18,27,40.5$). Linear or exponential? Model?

U3. A scatterplot is tight around a rising line except one point far below. The line of best fit

A) is pulled a little toward that point B) ignores outliers always C) becomes vertical D) must pass through that point

U4. Which is a linear model for “starts at 80 L and drains 2.5 L/min”?

A) ($V=80-2.5t$) B) ($V=80(0.025)^t$) C) ($V=2.5-80t$) D) ($V=80/2.5t$)

U5. (SPR) Using ($C=12n+45$), cost of 8 shirts.

Nitpick

- Slope units: “(y)-units per (x)-unit”
 - Don’t call an exponential table linear because it “goes up”
-

Block 2 — Probability and two-way tables (1:10–2:10) · C5

Oral

1. $P(A) = \frac{\text{count } A}{\text{total}}$
2. $P(A \mid B) =$ restrict to the **B** row/column first
3. “And” vs “or” (keep light: SAT mostly wants table reading)

Club table (same as Weekend 1 — she should be faster):

	Yes	No	Total
Junior	12	8	20
Senior	18	12	30
Total	30	20	50

Items

U6. $P(\text{Yes})$ for a random member?

U7. $P(\text{Yes} \mid \text{Senior})$?

U8. $P(\text{Junior} \mid \text{No})$?

U9. How many seniors said No?

U10. A spinner is fair with 5 equal sectors, two of them red. $P(\text{red})$? (SPR as a decimal or fraction)

U11. A bag: 4 red, 6 blue. One marble drawn. $P(\text{not red})$?

U12. (MC) From the club table, $P(\text{Senior and Yes})$ is

- A) $(18/30)$ B) $(18/50)$ C) $(30/50)$ D) $(18/20)$

Nitpick

- Conditional: new denominator
 - Joint “and”: over the **grand** total unless they already conditioned
-

Break (2:10–2:20)

Block 3 — Margin of error + study design (2:20–3:15) · C6 / C7

Oral

1. Sample percent 52%, MOE 4%. Interval?
2. Does MOE mean “the candidate wins”?
3. Larger random sample → MOE usually **smaller**.
4. Honors-hallway survey claiming “the school agrees” — what’s wrong?
5. Observational study vs experiment (random assignment).

Items

U13. 52% support, MOE 4%. Best conclusion?

- A) Exactly 52% of all voters support it
B) True support is likely between 48% and 56%
C) The candidate will win
D) The sample is biased because it isn’t 50–50

U14. Two random samples, same population. Sample A ($n=100$), Sample B ($n=900$). Which likely has the **smaller** margin of error?

- A) A B) B C) always equal D) cannot say without the percent

U15. A researcher asks only students in the honors hallway whether they like a new schedule, then claims the **whole school** supports it. The main problem is

A) the mean is undefined B) convenience / biased sample C) margin of error is 0 D) it is a well-designed experiment

U16. To test whether a new tutor program **causes** higher scores, you should

- A) survey whoever already bought the program
- B) randomly assign students to program vs not, then compare scores
- C) interview only A+ students
- D) use the median instead of the mean

U17. A news story says “people who eat more vegetables have lower blood pressure, so vegetables cause lower blood pressure.” The SAT-level issue:

A) percents cannot apply to health B) possible confounding (exercise, income, etc.) C) the median is larger D) MOE must be 50%

U18. A random sample of 400 students: 61% want later start, MOE 5%. Which is **not** justified?

- A) The interval 56%–66% is a reasonable range for the true percent
- B) More than half of all students might want a later start
- C) Exactly 61% of all students want a later start
- D) A larger sample would likely narrow the interval

Nitpick

- C6 is almost always **interval language**, not certainty
 - C7 is **who was asked** and **was anything randomly assigned**
-

Block 4 — Timed PSDA mix (3:15–3:50)

12 questions, 16 minutes, then review misses.

T1. Ratio 3:1 oil:vinegar, 2 cups vinegar. Cups of oil?

T2. 18 is 45% of what?

T3. \$80 up 25% then down 25%. Final?

T4. Data (2,2,3,7,16). Median?

- T5.** Five tests mean 80; four scores 72, 88, 90, 70. Fifth?
- T6.** ($y=8x+42$). Score at 6 hours?
- T7.** Club table: ($P(\text{Yes} \mid \text{Senior})$)?
- T8.** Club table: ($P(\text{Senior and Yes})$)?
- T9.** $52\% \pm 4\%$. Lower end of the interval?
- T10.** Honors-hallway survey of “the school.” Bias? (yes/no + one word)
- T11.** Combined printers 40+55 pages/min, 600 pages. Time?
- T12.** Population 800, two successive +10%. New population?
-

Block 5 — Recap + take-home (3:50–4:00)

1. practice-set-03.md, 16 items, **24 minutes**.
 2. Question Bank: PSDA **all skills**, 20 items — force some C6/C7, don’t only drill percents.
 3. Preview Weekend 4: Geometry & Trig + Module 1 pacing.
-

Instructor notes

- If tables are slow, redraw U6–U8 with her talking **denominator first**.
- C6/C7 should feel like English class with a tiny bit of math. Don’t overteach stats theory.

Weekend 3 practice set

16 questions, 24 minutes. Answers: answer-key-03.md.

1. **(C1, SPR)** Flour:sugar ($=2:5$). 12 cups flour. Cups of sugar?
2. **(C1, SPR)** 180 miles in 3 hours. Miles in 2.5 hours at the same rate?
3. **(C1, MC)** Concentrate:water ($=1:3$). Cups of concentrate in 16 cups of mix?
A) 3 B) 4 C) 5 D) 12

4. **(C2, MC)** 40 is what percent of 160?
A) 20% B) 25% C) 40% D) 4%
5. **(C2, SPR)** 18 is 45% of what number?
6. **(C2, MC)** \\$80 up 25% then down 25%. Final price?
A) \\$80 B) \\$75 C) \\$85 D) \\$70
7. **(C2, MC)** 800 grows 10%, then 10% again. New population?
A) 960 B) 968 C) 880 D) 1000
8. **(C2, SPR)** What number, increased by 20%, becomes 72?
9. **(C3, MC)** Data (2,2,3,7,16). Median?
A) 2 B) 3 C) 6 D) 7
10. **(C3, SPR)** Five tests, mean 80. Scores so far: 72, 88, 90, 70. Fifth score?
11. **(C3, MC)** Adding 100 to (2,2,3,7,16) will
A) raise the median more than the mean
B) raise the mean more than the median
C) leave both unchanged
D) lower the range
12. **(C4, SPR)** ($y=8x+42$). Predicted score at 6 hours?
13. **(C5, MC)** Club table (J 12Y/8N, S 18Y/12N). ($P(\text{Yes} \mid \text{Senior})$)
A) (18/50) B) (18/30) C) (30/50) D) (18/20)
14. **(C5, MC)** Same table. ($P(\text{Senior and Yes})$)
A) (18/30) B) (18/50) C) (30/50) D) (18/20)
15. **(C6, MC)** $52\% \pm 4\%$. Best conclusion?
A) Exactly 52% of all voters
B) True value likely between 48% and 56%
C) The candidate wins
D) The sample is biased because it isn't 50–50
16. **(C7, MC)** Honors-hallway survey used to speak for the whole school. Main problem?
A) mean undefined B) biased / convenience sample C) MOE is 0 D) it is a perfect

experiment

Weekend 4 overview

Goal: Close D1–D4 leaks, then run a Module 1-style mixed set and lock pacing rules.

Time: Saturday 4h + Sunday 4h

Day	File	Focus
Saturday	saturday.md	Area/volume, similar triangles/angles, Pythagorean + special rights
Sunday	sunday.md	Trig ratios, circles, 16-item Module 1 sim, adaptive clinic
Practice	practice-set-04.md	16-item take-home, 24 min
Answers	answer-key-04.md	Worked solutions

After this weekend: Official Bluebook Math module(s) + error log. No more new teaching packs unless a skill is still .

Weekend 4 Saturday

Theme: Geometry & Trig — formulas she already knows, SAT’s favorite twists

Skills: D1 area/volume · D2 lines/angles/similar triangles · D3 Pythagorean + special rights

Mode: Name the figure and the *scale type* before calculating.

All items are **instructor-created SAT-style**.

Block 0 — Geometry on SAT (0:00–0:15)

Oral

1. If lengths $\times 2$, area? volume?

2. Triangle angle sum.
3. Recite 45-45-90 and 30-60-90.
4. When do you *not* need a diagram number — just a ratio?

Whiteboard:

Scale on lengths	Area / surface	Volume
(k)	(k^2)	(k^3)

SAT will give a figure that looks familiar and then ask the **scaled** quantity.

Block 1 — Area and volume (0:15–1:15) • D1

Items

- S1.** Cube edge 3. Volume? If edge doubles, new volume?
- S2.** Rectangle 8 by 6. Area? Diagonal? (SPR for the diagonal)
- S3.** Triangle base 10, height 7. Area?
- S4.** A rectangular prism is 4 by 3 by 5. Volume? Surface area?
- S5.** Two similar paintings. Side lengths in ratio (2:3). If the smaller area is 20, the larger area is?
A) 30 B) 40 C) 45 D) 60
- S6.** A cylinder has radius 3 and height 4. Volume in terms of (π) ? (SPR: enter the coefficient of (π))
- S7.** Every length of a sculpture is scaled by $(\frac{1}{2})$. The new volume is what fraction of the original?
A) $(\frac{1}{2})$ B) $(\frac{1}{4})$ C) $(\frac{1}{8})$ D) $(\frac{1}{16})$

Nitpick

- Area uses **two** dimensions $((k^2))$, volume three $((k^3))$
- Cylinder: $(V = \pi r^2 h)$, not $(\pi r h)$

- Surface area: 2 of each pair of faces
-

Block 2 — Angles, parallel lines, similar triangles (1:15–2:15) · D2

Oral

1. Parallel lines + transversal: which angles are equal?
2. Similar triangles: equal **angles**, proportional **sides**.
3. Corresponding sides sit across from corresponding angles.

Items

S8. Triangle angles (40°) and (65°). Third angle?

S9. Two similar triangles. Corresponding sides 6 and 8. Side 9 on the smaller corresponds to what on the larger?

S10. Similar triangles, corresponding sides 4 and 10. Smaller perimeter 18. Larger perimeter?

S11. Parallel lines cut by a transversal. One interior angle is (110°). The alternate interior angle is?

A) (70°) B) (110°) C) (20°) D) (180°)

S12. A triangle has angles (x), ($2x$), and ($3x$). Find (x). (SPR)

S13. An isosceles triangle has vertex angle (40°). A base angle is?

S14. (MC) Similar figures, scale factor (3). A 5° angle on the small figure becomes

A) (15°) B) (5°) C) (45°) D) (3°)

Nitpick

- **Angles do not scale.** Similarity copies angles.
 - Perimeter scales by (k); area by (k^2).
-

Break (2:15–2:25)

Block 3 — Pythagorean and special rights (2:25–3:35) · D3

Oral

1. When can you skip $a^2+b^2=c^2$ and just use a pattern?
2. 45-45-90: legs (s), hyp ($s\sqrt{2}$).
3. 30-60-90: short (s), long ($s\sqrt{3}$), hyp (2s).
4. Short leg is opposite 30° .

Items

S15. Legs 5 and 12. Hypotenuse?

S16. Isosceles right triangle, leg 5. Hypotenuse?

S17. 30-60-90, short leg 6. Long leg and hypotenuse?

S18. Rectangle 5 by 12. Diagonal?

S19. A right triangle has hyp 10 and one leg 6. The other leg? (SPR)

S20. 45-45-90, hypotenuse ($6\sqrt{2}$). A leg is?

A) 6 B) 3 C) ($6\sqrt{2}$) D) 12

S21. 30-60-90, hypotenuse 14. Short leg?

A) 7 B) ($7\sqrt{3}$) C) 14 D) ($14\sqrt{3}$)

S22. (SPR) Right triangle legs 9 and 12. Hypotenuse. (Recognize the 3-4-5 multiple.)

Nitpick

- 6-8-10 and 9-12-15 are 3-4-5
- 5-12-13 and 9-12-15 are different families — don't mash them
- Hypotenuse is **longest**; don't put (2s) on a leg

Block 4 — Mixed D1–D3 sprint (3:35–3:55)

M1. Area of a circle radius 4, in terms of π .

M2. Third angle if two angles are 90° and 35° .

M3. Cube edge 2 $\rightarrow$ volume?

M4. Scale lengths $\times 3 \rightarrow$ area $\times ?$

M5. 45-45-90 leg 7 $\rightarrow$ hyp?

Block 5 — Exit + homework (3:55–4:00)

E1. Similar sides 2:5. Small area 8. Large area?

E2. Angles do / do not scale under similarity.

E3. 30-60-90 short leg 5. Hypotenuse?

Assign

Question Bank: Geometry and Trig, 15 items. Sunday adds trig ratios + circles + a Module 1 simulation.

Weekend 4 Sunday

Theme: Right-triangle trig, circles, then a Module 1-style mix + strategy

Skills: D3 trig · D4 circles · mixed A/B/C/D pacing

Block 0 — Saturday miss review (0:00–0:15)

Re-work every D miss. If scale factors leaked, redo S5 and S7 cold.

Block 1 — Right-triangle trig (0:15–1:10) · D3

Oral

1. SOH-CAH-TOA, out loud.
2. ($\sin \theta = \cos(90^\circ - \theta)$). Why does SAT like that?
3. You know ($\sin \theta = 3/5$). A possible opposite and hypotenuse? Need the third side?

Items

U1. Opposite 3, hypotenuse 5. ($\sin \theta =$) ?

U2. Adjacent 8, hypotenuse 10. ($\cos \theta =$) ?

U3. Opposite 3, adjacent 4. ($\tan \theta =$) ? Hypotenuse?

U4. ($\sin \theta = \frac{5}{13}$). ($\cos \theta =$) ? (right triangle, acute (θ))

U5. (SPR) In a right triangle, ($\tan \theta = \frac{3}{4}$). Enter ($\sin \theta$) as a decimal.

U6. Which is equal to ($\cos 20^\circ$)?

- A) ($\sin 20^\circ$) B) ($\sin 70^\circ$) C) ($\tan 20^\circ$) D) ($\cos 70^\circ$)

Nitpick

- Label opposite/adjacent **relative to the angle named**
 - Pythagorean to finish the third side before finding the other ratio
-

Block 2 — Circles (1:10–2:00) · D4

Oral

1. ($C = 2\pi r = \pi d$). ($A = \pi r^2$).
2. Arc length ($= \frac{\theta}{360} \cdot 2\pi r$).
3. Sector area ($= \frac{\theta}{360} \cdot \pi r^2$).
4. Circle equation ($(x-h)^2 + (y-k)^2 = r^2$). Center and radius?

Items

U7. Diameter 10. Circumference in terms of (π) ? (SPR: coefficient of (π))

U8. Radius 4. Area in terms of (π) ?

U9. Radius 6, central angle 60° . Arc length?

U10. Same circle, 60° sector area in terms of (π) ? (SPR: coefficient of (π))

U11. $((x-2)^2+(y+1)^2=16)$. Center and radius?

U12. A central angle of 90° on a circle of radius 8. Arc length in terms of (π) ?

Nitpick

- Diameter vs radius (U7 is the classic rush miss)
 - 60° is $(\frac{1}{6})$ of the circle, 90° is $(\frac{1}{4})$
-

Break (2:00–2:10)

Block 3 — Module 1 simulation (2:10–3:00)

16 questions, 25 minutes. Mixed domains, easier $\rightarrow$ medium (Module 1 feel). Silent. Flag at 2:30. Desmos allowed.

M1. (A) $(5-2(3x+1)=x-17)$

M2. (A) Slope of $(5x+2y=10)$

M3. (A) $(\begin{cases} x+y=10 \\ 2x-y=5 \end{cases})$ — the (x) -value

M4. (B) Zeros of $(2(x-3)(x+1))$

M5. (B) Vertex (x) -coordinate of $(y=x^2-8x+10)$

M6. (B) $(2^{\{x+1\}}=16)$

M7. (C) 18 is 45% of what?

M8. (C) \$80 up 25% then down 25%. Final?

M9. (C) Club table: $(P(\text{Yes} \mid \text{Senior}))$ (Juniors 12Y/8N, Seniors 18Y/12N)

M10. (C) $52\% \pm 4\%$. Lower end of the likely interval?

M11. (D) Third angle if 40° and 65°

M12. (D) Legs 5 and 12. Hypotenuse?

M13. (D) Cube edge 3. Volume?

M14. (D, SPR) Opposite 3, hypotenuse 5. Enter $(\sin \theta)$ as a decimal.

M15. (B) $(|x-2|=-5)$. Number of solutions?

M16. (A) $(-2x+5>13)$. Solution set in words or inequality.

After the timer: 10 minutes — only review hesitations and misses. Student explains first.

Block 4 — Adaptive / pacing clinic (3:10–3:50)

No new content. Oral + two “decision” reps.

Decision rules (write on the board)

1. **Module 1 is a routing test.** Two careless Algebra misses can cap the 780.
2. First pass: bank every A and easy C/D.
3. Desmos when: systems, “which graph,” intersections, “equivalent” check.
4. Hand algebra when: one-step isolate, percent, table probability.
5. 2:30 stuck → flag, guess if needed, move. Come back if time.
6. SPR: never leave blank.
7. Last 60 seconds: any blank SPR + any unanswered MC.

Two pacing reps (talk, don’t fully solve unless useful)

R1. A long exponential word problem appears as question 6 of 22. Do you grind it now?

R2. Question 20 is a circle sector and you have 4 minutes + two unanswered Algebra SPRs. Order of operations?

Error-log standard from here on

Every Bluebook / Question Bank miss: skill ID + (concept / setup / arithmetic / rushed / wording) + “would Desmos have caught it?”

Block 5 — Recap + take-home (3:50–4:00)

1. practice-set-04.md, 16 items, **24 minutes** (D-heavy + a few mixed).
 2. **Homework that is not in this repo:** one Bluebook Math section (or both Math modules if energy). Log every miss on ../01-math-content-map.md.
 3. After Weekend 4: Weekends 5+ are official timed practice + clinic, not new teaching packs.
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Instructor notes

- If trig ratios are automatic, steal time for extra circle + scale-factor items — those are the remaining freebies.
- The simulation score is **not** a 200–800 conversion. Use it as “Module 1 accuracy.” Target: ≤ 2 misses on M1–M16.

Weekend 4 practice set

16 questions, 24 minutes. Answers: answer-key-04.md.

1. **(D1, SPR)** Volume of a cube with edge 3.
2. **(D1, MC)** Lengths scaled by $\left(\frac{1}{2}\right)$. New volume is what fraction of the original?
A) $\left(\frac{1}{2}\right)$ B) $\left(\frac{1}{4}\right)$ C) $\left(\frac{1}{8}\right)$ D) $\left(\frac{1}{16}\right)$
3. **(D1, MC)** Similar paintings, sides (2:3). Smaller area 20. Larger area?
A) 30 B) 40 C) 45 D) 60
4. **(D2, MC)** Angles (40°) and (65°) . Third angle?
A) (65°) B) (75°) C) (85°) D) (95°)
5. **(D2, SPR)** Triangle angles (x) , $(2x)$, $(3x)$. Enter (x) .
6. **(D2, MC)** Similar figures, scale factor 3. A (5°) angle becomes
A) (15°) B) (5°) C) (45°) D) (3°)

7. **(D3, SPR)** Legs 5 and 12. Hypotenuse?
8. **(D3, MC)** 45-45-90, leg 5. Hypotenuse?
A) 5 B) $(5\sqrt{2})$ C) $(5\sqrt{3})$ D) 10
9. **(D3, MC)** 30-60-90, hypotenuse 14. Short leg?
A) 7 B) $(7\sqrt{3})$ C) 14 D) $(14\sqrt{3})$
10. **(D3, MC)** Opposite 3, hypotenuse 5. $(\sin\theta=)$
A) $(3/4)$ B) $(3/5)$ C) $(4/5)$ D) $(4/3)$
11. **(D3, MC)** $(\cos 20^\circ=)$
A) $(\sin 20^\circ)$ B) $(\sin 70^\circ)$ C) $(\tan 20^\circ)$ D) $(\cos 70^\circ)$
12. **(D4, SPR)** Diameter 10. Enter the coefficient of (π) in the circumference.
13. **(D4, MC)** Radius 6, arc 60° . Arc length?
A) (π) B) (2π) C) (6π) D) (12π)
14. **(D4, MC)** $((x-2)^2+(y+1)^2=16)$. Radius?
A) 2 B) 4 C) 8 D) 16
15. **(A5, MC)** $(-2x+5>13)$
A) $(x>-4)$ B) $(x<-4)$ C) $(x>4)$ D) $(x<4)$
16. **(C2, MC)** \$80 up 25% then down 25%. Final?
A) \$80 B) \$75 C) \$85 D) \$70

Answer keys

Instructor side. Do not hand these pages to Karina during a timed set.

Answer key — Weekend 1

Worked solutions for Saturday survey items, Sunday diagnostic + stations, exit ticket, Practice Set 01, and the optional Algebra drill.

Verify SPRs by substitution.

Saturday — Algebra (S1–S6)

S1. $(5-2(3x+1)=x-17)$

$(5-6x-2=x-17) \rightarrow (3-6x=x-17) \rightarrow (3+17=7x) \rightarrow (x=\frac{20}{7})$.

S2. LCD 12: $(3x+2x=60) \rightarrow (x=12)$.

S3. Second $\times 2$: $(2x-4y=10)$, first is $(2x-4y=6) \rightarrow$ parallel distinct $\rightarrow$ **0 solutions**.

S4. $(-2x>8) \rightarrow$ flip $\rightarrow (x<-4)$. **B**.

S5. $(3x-2y=12) \rightarrow (y=\frac{3}{2}x-6)$. Slope $(\frac{3}{2})$; (y)-intercept (-6) .

S6. $(40+25m=165) \rightarrow (m=5)$.

Saturday — Advanced Math (S7–S12)

S7. Factor $((x-3))$: $(4x(x-3)-2(x-3)=(x-3)(4x-2)=2(x-3)(2x-1))$. A and C are the same. **D**.

S8. Vertex $((-1,5))$; $(a=-2<0) \rightarrow$ **maximum**.

S9. Factor between 0 and 1: **B** ($y=20(0.85)^x$).

S10. ($2x-3=7$) $\rightarrow$ ($x=5$); ($2x-3=-7$) $\rightarrow$ ($x=-2$). Solutions: ($x=5$) or ($x=-2$).

S11. ($x^2=x+2$) $\rightarrow$ ($x^2-x-2=0$) $\rightarrow$ ($(x-2)(x+1)=0$) $\rightarrow$ **2 solutions**.

S12. Axis ($x=2$); ($y=4-8-1=-5$). Minimum -5 .

Saturday — PSDA (S13–S18)

S13. ($\frac{2}{5}=\frac{12}{s}$) $\rightarrow$ ($2s=60$) $\rightarrow$ ($s=30$) cups.

S14. ($80 \times 1.25=100$), then ($100 \times 0.75=75$). Final **\$75**.

S15. Mean ($=(2+2+3+7+16)/5=6$). Median ($=3$). The **mean** is pulled up by 16.

S16. Slope 8 = points per study hour. ($y=8(6)+42=90$).

S17. Restrict to seniors: ($18/30=\frac{3}{5}=0.6$).

S18. B — a margin of error of 4% around 52% means the true value is likely between 48% and 56%. A is over-precise; C is a prediction; D is not what MOE says.

Saturday — Geometry (S19–S24)

S19. ($3^3=27$). Doubled edge $\rightarrow$ scale ($k=2$) $\rightarrow$ volume ($\times 8$) $\rightarrow$ (216).

S20. ($\frac{6}{8}=\frac{9}{x}$) $\rightarrow$ ($6x=72$) $\rightarrow$ ($x=12$).

S21. 45-45-90: hypotenuse ($=5\sqrt{2}$).

S22. ($\sin\theta=\frac{3}{5}$).

S23. Circumference (12π); ($60/360=\frac{1}{6}$) $\rightarrow$ arc ($=2\pi$).

S24. ($6^2+8^2=36+64=100$) $\rightarrow$ diagonal **10**.

Saturday — Exit ticket

E1. $(4x-4-x=2x+5) \rightarrow (3x-4=2x+5) \rightarrow (x=9)$.

E2. Zeros $(x=3)$ and $(x=-1)$.

E3. $(\frac{40}{160}=\frac{1}{4}=25\%)$.

E4. $(\pi r^2=16\pi)$.

Sunday — Diagnostic (D1–D12)

D1. $(7-6x+3=2x+8) \rightarrow (10-6x=2x+8) \rightarrow (2=8x) \rightarrow (x=\frac{1}{4})$.

D2. Add: $(3x=15) \rightarrow (x=5), (y=5)$.

D3. $(-\frac{1}{2}x \leq 4) \rightarrow$ multiply by (-2) (flip): $(x \geq -8)$.

D4. $((x-7)(x+2))$. Smaller constant -7 .

D5. $(x=-\frac{b}{2a}=\frac{8}{2}=4)$.

D6. $(200 \times (1.5)^2=450)$.

D7. Rate $(60 \text{ mph}) \rightarrow (60 \times 2.5=150)$.

D8. Sale $(60 \times 0.85=51)$; tax $(51 \times 1.08=55.08)$.

D9. $(30/50=\frac{3}{5}=0.6)$.

D10. $(5^2+12^2=25+144=169=13^2)$. Hypotenuse **13**.

D11. $(180-40-65=75^\circ)$.

D12. Diameter 10 $\rightarrow (C=10\pi)$.

Sunday — Stations (U1–U12)

U1. $(35+0.10t=47) \rightarrow (t=120)$ texts.

U2. Perpendicular to slope 3 $\rightarrow$ slope $(-\frac{1}{3})$; through $((0,4)) \rightarrow$ **B**.

U3. $(a+c=15), (12a+7c=145). (c=15-a): (12a+7(15-a)=145) \rightarrow (5a=40) \rightarrow (a=8).$

U4. $(x^2-8x+16).$

U5. $(-16t^2+32t+5=0) \rightarrow (16t^2-32t-5=0) \rightarrow$

$(t=\frac{32\pm\sqrt{1024+320}}{32}=\frac{32\pm\sqrt{1344}}{32}=\frac{32\pm 8\sqrt{21}}{32}=\frac{4\pm\sqrt{21}}{4}).$

Positive root: $(\frac{4+\sqrt{21}}{4})$ seconds.

U6. Ratios $(1.5,1.5,1.5) \rightarrow$ exponential $(y=12(1.5)^x).$

U7. $(0.45x=18) \rightarrow (x=40).$

U8. Mean $(=(70+74+74+76+92)/5=77.2).$ Median $(=74).$ Quote the **median**.

U9. Convenience / biased sample — honors hallway is not the whole school (C7).

U10. Scale $(10/4=2.5) \rightarrow$ perimeter $(18\times 2.5=45).$

U11. Long leg $(6\sqrt{3});$ hypotenuse $(12).$

U12. $(\cos\theta=\frac{8}{10}=\frac{4}{5}).$

Sunday — SPR drill (P1–P5)

P1. $\times 10: (2(x+15)=5x-30) \rightarrow (2x+30=5x-30) \rightarrow (x=20).$

P2. $(120-4t=0) \rightarrow (t=30).$

P3. 3 doublings $\rightarrow (100\cdot 8=800).$

P4. $(5^2+12^2=169) \rightarrow$ **13**.

P5. $(0.20\times 45=9).$

Practice Set 01

1. $(7-3(2x-1)=2x+8) \rightarrow (7-6x+3=2x+8) \rightarrow (10-6x=2x+8) \rightarrow (2=8x) \rightarrow (x=\frac{1}{4}).$ **A**.

2. $(2(5x-2)=4(x+6)) \rightarrow (10x-4=4x+24) \rightarrow (6x=28) \rightarrow (x=\frac{14}{3})$.
 3. $(2y=-5x+10) \rightarrow (y=-\frac{5}{2}x+5)$. Slope **A** $(-\frac{5}{2})$.
 4. Add: $(3x=15) \rightarrow (x=5), (y=5)$. **A**.
 5. Factor $((x+4))$: $(2x(x+4)-1(x+4)=(x+4)(2x-1))$. A and C are the same. **D**.
 6. Zeros $(x=3), (x=-1)$. **A**.
 7. $(x=-\frac{b}{2a}=4)$.
 8. **C** $(y=9(1.2)^x)$.
 9. $(3:1)$ with 2 cups vinegar $\rightarrow$ **6** cups oil.
 10. $(80 \times 1.25=100), (100 \times 0.75=75)$. **B**.
 11. Ordered already; middle value **3**. **B**. (Mean is 6; don't confuse them.)
 12. Seniors only: $(18/30)$. **B**.
 13. (27) .
 14. $(180-40-65=75)$. **B**.
 15. **B** $(5\sqrt{2})$.
 16. $(C=\pi d=10\pi) \rightarrow$ enter **10**.
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Optional Algebra drill

Items 1–16 match the original Saturday Algebra set.

1. $(x=\frac{20}{7})$
2. $(x=12)$
3. $(x=20)$
4. $(n-18=4n) \rightarrow (n=-6)$
5. $(x=8)$

6. $(y=\frac{3}{2}x-6)$; slope $(\frac{3}{2})$; intercept (-6)
7. $(y-7=-\frac{1}{2}(x+4)) \rightarrow (y=-\frac{1}{2}x+5)$
8. Slope 12 = $\$/shirt$; intercept 45 = fixed fee; $(C=141)$
9. **B**
10. $(x=\frac{32}{7})$, $(y=\frac{43}{7})$
11. $(x=\frac{13}{10})$, $(y=\frac{8}{5})$
12. **0 solutions**
13. **8** adult tickets
14. $(x>-3)$
15. **B** $((1,2))$
16. $(k>4)$

Answer key — Weekend 2

Saturday

- S1.** $(\frac{(x-3)(x+3)}{x-3}=x+3)$. **B.**
- S2.** $(\frac{2x(x^2-4)}{2x}=x^2-4)$.
- S3.** Difference of cubes $(a=2x)$, $(b=3)$: $((2x-3)(4x^2+6x+9))$.
- S4.** $((x-3)(4x-2)=2(x-3)(2x-1))$. **A.**
 (C is $(4x^2-10x+6)$, not the expansion $(4x^2-14x+6)$.)
- S5.** $((3x^2y)^2=9x^4y^2) \rightarrow (a+b+c=9+4+2=15)$.
- S6.** $(\sqrt{18x^2}=x\sqrt{18}=3x\sqrt{2})$ for $(x>0)$.
- S7.** $(3x+1=10) \rightarrow (x=3)$; $(3x+1=-10) \rightarrow (x=-\frac{11}{3})$.
- S8.** $(-3<x-4<3) \rightarrow (1<x<7)$.
- S9.** $(2x-5=x+1) \rightarrow (x=6)$; $(2x-5=-(x+1)) \rightarrow (2x-5=-x-1) \rightarrow (3x=4) \rightarrow (x=\frac{4}{3})$.
- S10.** Absolute value cannot equal a negative. **A** (0 solutions).

S11. $(2x-3=\pm 7) \rightarrow (x=5) \text{ or } (x=-2)$. Larger **5**.

S12. $(x+3=2(x-1)=2x-2) \rightarrow (5=x)$. Not 1, so valid.

S13. Multiply by $(x-2)$: $(1+3(x-2)=5) \rightarrow (1+3x-6=5) \rightarrow (3x=10) \rightarrow (x=\frac{10}{3})$.

S14. $(\frac{x-2}{x-1}=1) \rightarrow (x-2=x-1) \rightarrow (-2=-1)$, contradiction. $(x=1)$ forbidden. **No solution**.

S15. Undefined at $(x=2)$. Numerator $((x-2)(x-3)) \rightarrow$ simplified $(x-3)$.

S16. $(\frac{4}{x}=3-\frac{12}{5}) \rightarrow (x=\frac{8}{5})$.

S17. $(2x+3=x^2) \rightarrow (x^2-2x-3=0) \rightarrow (x=3) \text{ or } (-1)$.

$(x=-1)$: $(\sqrt{1}=1 \neq -1)$. **Only $(x=3)$** .

S18. $(x+12=x^2) \rightarrow (x^2-x-12=0) \rightarrow (x=4) \text{ or } (-3)$. Only **$(x=4)$** .

S19. $(3x-2=16) \rightarrow (x=6)$. Check: $(\sqrt{16}=4)$. ✓

S20. $(5x+1=(x+1)^2=x^2+2x+1) \rightarrow (x^2-3x=0) \rightarrow (x=0) \text{ or } (3)$. Both check. Larger **3**.

E1. $(x+4)$.

E2. $(x=4) \text{ or } (x=-14)$.

E3. No. $(\sqrt{1}=1)$ but $(1-2=-1)$. Also $(\sqrt{\cdot})$ cannot equal a negative.

Sunday — Blocks 1–3

U1. $(2^{x+1}=2^4) \rightarrow (x=3)$.

U2. $(3^x=9=3^2) \rightarrow (x=2)$.

U3. $(\left(\frac{12}{t}\right)^t=\frac{14}{\left(\frac{12}{t}\right)^2}) \rightarrow (t=2)$.

U4. C.

U5. 3 periods: $(80 \cdot 3^3=2160)$.

U6. $(V(t)=20000(0.85)^t)$; $(V(2)=20000 \cdot 0.7225=14450)$.

U7. $(x^2=2x+3) \rightarrow (x^2-2x-3=0) \rightarrow ((x-3)(x+1)=0) \rightarrow ((3,9), (-1,1))$.

U8. $(x^2-x+1=0)$, disc $(1-4<0)$. **A** (0).

U9. $(x+x^2-1=5) \rightarrow (x^2+x-6=0) \rightarrow ((2,3)) \text{ and } ((-3,8))$.

U10. $(x^2 - x - 4 = 0) \rightarrow (x = \frac{1 + \sqrt{17}}{2})$ (the larger).

U11. $(x^2 + 9 = 25) \rightarrow (x = \pm 4)$. Points $((4, 3))$ and $((-4, 3))$. **Two.**

U12. Vertex $((1, 5))$, **minimum**, $(f(3) = 13)$.

U13. $(g(0) = 3)$, $(g(4) = 48)$. 3 = initial value; 2 = growth factor (doubles each step).

U14. $(t = \frac{b}{2a} = \frac{48}{32} = 1.5)$; $(h(1.5) = 40)$.

U15. **B** (constant *factor*, intercept 7).

U16. **9.**

Sunday — Timed mix

T1. $(x = -4, 3)$.

T2. $(x^2 - 8x + 16)$.

T3. $(2x - 3)$.

T4. $(x = 4)$ or $(x = -14)$.

T5. $(x = 5)$.

T6. $(x = 6)$.

T7. $(x = 4)$.

T8. **2.**

T9. $((-1, 5))$.

T10. **450.**

T11. **-5.**

T12. $(y = 20(0.85)^x)$.

Practice Set 02

1. **B** $(x + 3)$

2. **15**

3. **B**

4. **B** (3) and $(-\frac{11}{3})$

5. **B** $(1 < x < 7)$

6. **6**

- 7. **A** 0
- 8. **5**
- 9. **C** no solution
- 10. **B** 3 only
- 11. **6**
- 12. **B** ($x=3$)
- 13. **2160**
- 14. **C** 2
- 15. **B** min 5 at ($x=1$)
- 16. **9**

Answer key — Weekend 3

Saturday

S1. $(\frac{2}{5} = \frac{12}{s}) \rightarrow (s=30)$.

S2. $60 \text{ mph} \times 2.5 \text{ h} = \mathbf{150}$.

S3. Combined 95 pages/min $\rightarrow (600/95 = \frac{120}{19})$ minutes (≈ 6.32).

S4. $(\frac{3}{8} = \frac{x}{20}) \rightarrow (x = \frac{15}{2} = 7.5)$.

S5. $(1.5 \times 48 = 72)$ miles.

S6. 4 equal parts; $(16/4 = 4)$ cups concentrate.

S7. A: $(5 \text{ miles} / (0.2 \text{ h}) = \mathbf{25 \text{ mph}}$. B 20. C and D are 24 mph. **A**.

S8. $(40/160 = 25\%)$.

S9. $(0.45x = 18) \rightarrow (x = 40)$.

S10. $(80 \times 1.25 = 100)$, $(100 \times 0.75 = 75)$.

S11. $(60 \times 0.85 = 51)$, $(51 \times 1.08 = 55.08)$.

S12. ($\frac{15}{50}=30\%$).

S13. ($800 \times 1.1 \times 1.1=968$). **B.**

S14. ($1.2x=72$) $\rightarrow$ ($x=60$).

S15. Mean 6; median 3; range ($16-2=14$).

S16. Mean 77.2; median 74. Quote the **median**.

S17. Sum must be 400. Known sum 320. Fifth **80**.

S18. Both medians 5; means (6) vs (10). **B.**

S19. New median of six values is 5 (was 3); mean jumps from 6 to ($130/6 \approx 21.7$). **B.**

S20. Right tail pulls the mean up. **B.**

M1. 15. **M2.** 80. **M3.** 12. **M4.** ($28-5=23$). **M5.** 132.

E1. ($50 \times 1.2=60$), ($60 \times 0.8=48$).

E2. 7 parts; ($28/7=4$); larger ($5 \times 4=20$).

E3. Sorted (3,3,7,9,11); median 7.

Sunday

U1. Slope = 8 points per hour; ($y=90$).

U2. Ratios 1.5 $\rightarrow$ ($y=12(1.5)^x$).

U3. **A.**

U4. **A.**

U5. ($12 \cdot 8+45=141$).

U6. ($30/50=0.6$).

U7. ($18/30=0.6$).

U8. No row: 20 people; juniors among them 8 $\rightarrow$ ($8/20=0.4$).

U9. **12.**

U10. ($2/5=0.4$).

U11. $(6/10=0.6)$.

U12. B $(18/50)$.

U13. B.

U14. B (larger (n) , smaller MOE).

U15. B.

U16. B.

U17. B.

U18. C (exact 61% is not justified).

Sunday timed mix

T1. 6. **T2.** 40. **T3.** 75. **T4.** 3. **T5.** 80. **T6.** 90.

T7. $(18/30)$. **T8.** $(18/50)$. **T9.** 48. **T10.** Yes — convenience/bias.

T11. $(\frac{120}{19})$ min. **T12.** 968.

Practice Set 03

1. **30**

2. **150**

3. **B** 4

4. **B** 25%

5. **40**

6. **B** \$75

7. **B** 968

8. **60**

9. **B** 3

10. **80**

11. **B**

12. **90**

13. **B** $(18/30)$

14. **B** $(18/50)$

15. **B**

16. **B**

Answer key — Weekend 4

Saturday

S1. (27); doubled edge $\rightarrow (\times 8) \rightarrow (216)$.

S2. Area (48). Diagonal (10).

S3. $(\tfrac{12}{10} \cdot 10 \cdot 7 = 35)$.

S4. Volume (60). Surface $(2(4 \cdot 3 + 4 \cdot 5 + 3 \cdot 5) = 2(12 + 20 + 15) = 94)$.

S5. Area ratio (4:9). $(20 \cdot \tfrac{94}{4} = 470)$. **C**.

S6. $(V = \pi \cdot 9 \cdot 4 = 36\pi)$. Enter **36**.

S7. $(\left(\tfrac{12}{10}\right)^3 = \tfrac{18}{125})$. **C**.

S8. (75°) .

S9. $(\tfrac{6}{8} = \tfrac{9}{x}) \rightarrow (x = 12)$.

S10. Scale (2.5) $\rightarrow$ perimeter (45).

S11. Alternate interior angles equal. **B** (110°) .

S12. $(6x = 180) \rightarrow (x = 30)$.

S13. Base angles $(\tfrac{140}{2} = 70^\circ)$.

S14. Angles do not scale. **B** (5°) .

S15. (13).

S16. $(5\sqrt{2})$.

S17. Long ($6\sqrt{3}$); hyp (12).

S18. (13).

S19. (8) (6-8-10).

S20. A 6.

S21. Short (=) hyp($/2=7$). **A.**

S22. (15) (9-12-15).

M1. (16π). **M2.** (55°). **M3.** (8). **M4.** (9). **M5.** ($7\sqrt{2}$).

E1. ($k=\frac{52}{9}$), area ($\times \frac{25}{4}$) $\rightarrow$ ($8 \cdot \frac{25}{4}=50$).

E2. Do **not**.

E3. (10).

Sunday

U1. ($\frac{3}{5}$).

U2. ($\frac{4}{5}$).

U3. ($\tan=\frac{3}{4}$); hyp (5).

U4. 5-12-13 $\rightarrow$ ($\cos\theta=\frac{12}{13}$).

U5. 3-4-5 $\rightarrow$ ($\sin\theta=0.6$).

U6. B ($\sin 70^\circ$).

U7. (10).

U8. (16π).

U9. ($\frac{1}{6} \cdot 12\pi=2\pi$).

U10. ($\frac{1}{6} \cdot 36\pi=6\pi$). Enter **6**.

U11. Center ($(2,-1)$), radius (4).

U12. ($\frac{14}{3} \cdot 16\pi=4\pi$).

Module 1 simulation

M1. $(x=\frac{20}{7})$

M2. $(-\frac{5}{2})$

M3. $(x=5)$

M4. (3) and (-1)

M5. (4)

M6. (3)

M7. (40)

M8. (75)

M9. $(18/30)$

M10. (48)

M11. (75°)

M12. (13)

M13. (27)

M14. (0.6)

M15. (0)

M16. $(x<-4)$

Practice Set 04

1. **27**

2. **C** $(\frac{1}{8})$

3. **C** 45

4. **B** (75°)

5. **30**

6. **B** (5°)

7. **13**

8. **B** $(5\sqrt{2})$

9. **A** 7

10. **B** $(\frac{3}{5})$

11. **B** $(\sin 70^\circ)$

12. **10**

13. **B** (2π)

14. **B** 4

15. **B** ($x < -4$)

16. **B** \$75